Unit

35 Hess's Law and its applications

Unit Key Concepts

- Hess's Law
- Enthalpy change cycle

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

35.1 Hess's Law

Many compounds cannot be made directly from their constituent elements. In some cases, the reaction proceeds too slowly, or side reactions produce substances other than the desired compound. In all these cases, we determine the enthalpy changes of formation of the compounds by an indirect approach, which is based on **Hess's Law**.

Animation

Hess's Law

Hess's Law states:

✓ The enthalpy change of a reaction depends on the initial and final states of the reaction and is independent of the route by which the reaction may occur.

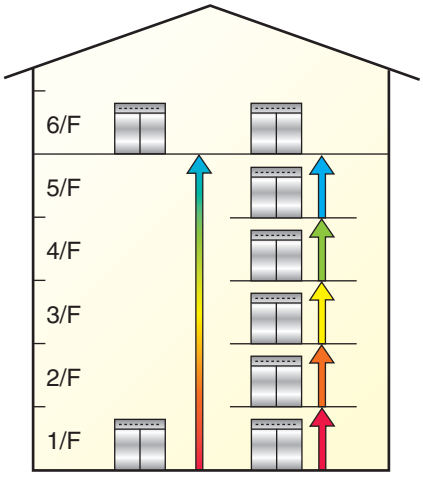


Fig. 35.1 An analogy for Hess's Law

Below is an analogy for Hess's Law (Fig. 35.1). Suppose you take the lift to go from the first floor to the sixth floor of a building. The total gain in your potential energy (which corresponds to the enthalpy change of the overall reaction) is the same whether you go directly there or stop at each floor on your way up (breaking the trip into a series of steps).

On the other hand, when you go down from the sixth floor to the first floor, the total loss in your potential energy (which corresponds to the enthalpy change of the overall reaction) is the same whether you go directly there or stop at each floor on your way down (breaking the trip into a series of steps).

Consider the reaction in which gaseous nitrogen and oxygen combine to form nitrogen dioxide gas. There are two ways we might get from the reactants to the product. The reaction could occur in a single step. The thermochemical equation for this single-step route is:

$$\frac{1}{2}N_2(g) + O_2(g) \longrightarrow NO_2(g) \qquad \Delta H_1 = +34 \text{ kJ mol}^{-1}$$

However, nitrogen dioxide can also be made using a two-step route. The thermochemical equations for the two steps are:

$$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \longrightarrow NO(g) \qquad \Delta H_2 = +90 \text{ kJ mol}^{-1}$$

$$NO(g) + \frac{1}{2}O_2(g) \longrightarrow NO_2(g) \qquad \Delta H_3 = -56 \text{ kJ mol}^{-1}$$

If we add these two equations, we get the equation for the overall reaction for the formation of nitrogen dioxide gas.

$$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \longrightarrow \cancel{NO(g)} \qquad \Delta H_2 = +90 \text{ kJ mol}^{-1}$$

$$\cancel{NO(g)} + \frac{1}{2}O_2(g) \longrightarrow NO_2(g) \qquad \Delta H_3 = -56 \text{ kJ mol}^{-1}$$

Overall reaction: $\frac{1}{2}N_2(g) + O_2(g) \longrightarrow NO_2(g) \qquad \Delta H_2 + \Delta H_3 = +34 \text{ kJ mol}^{-1}$

Notice that $\Delta H_1 = \Delta H_2 + \Delta H_3$

i.e. enthalpy change of the process in the single-step route
= sum of enthalpy changes of processes in the two-step route

The enthalpy change of the above reaction is the same whether it is carried out via the single-step route or the two-step route.

Enthalpy change cycle

We can use an **enthalpy change cycle** (Fig. 35.2) to relate the enthalpy changes of processes involved in the reaction between gaseous nitrogen and oxygen to form nitrogen dioxide gas.

enthalpy change cycle 焓變循環

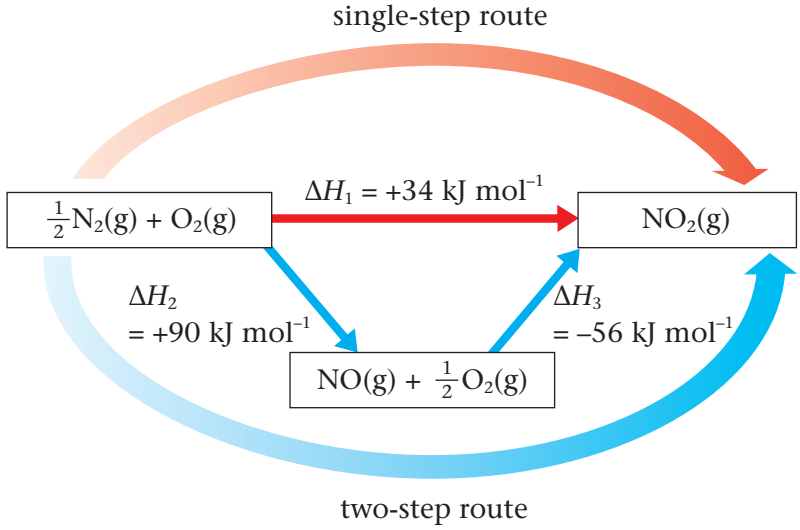


Fig. 35.2 An enthalpy change cycle relating the enthalpy changes of processes involved in the reaction between gaseous nitrogen and oxygen to form nitrogen dioxide gas

Example 35.1

Q Ethyne ($C_2H_2(g)$) can be converted to ethane ($C_2H_6(g)$) by two different routes.

Route 1

$C_2H_2(g) + 2H_2(g) \longrightarrow C_2H_6(g)$ $\quad \Delta H_1 = -313$ kJ mol^{-1}

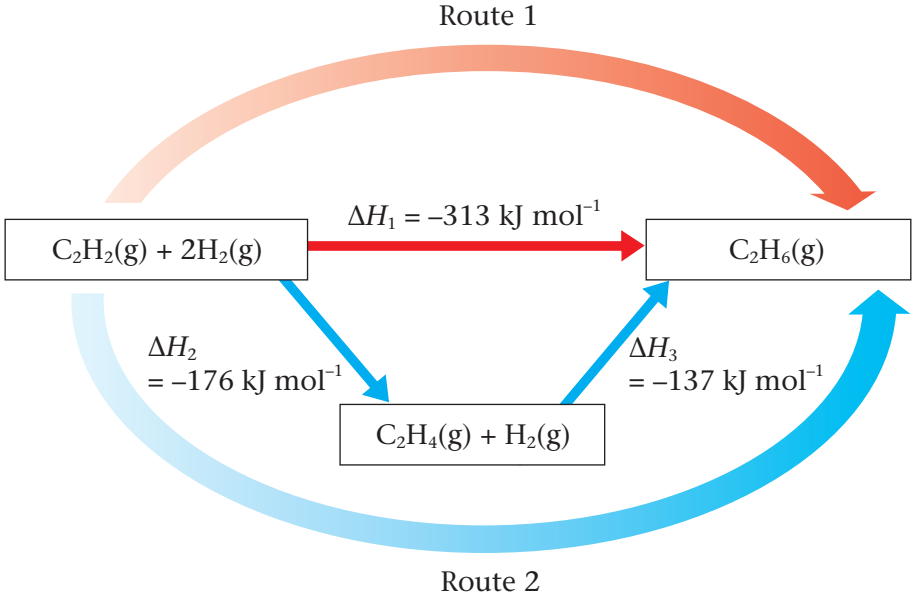
Route 2 (via the intermediate $C_2H_4(g)$)

$C_2H_2(g) + H_2(g) \longrightarrow C_2H_4(g)$ $\quad \Delta H_2 = -176$ kJ mol^{-1}

$C_2H_4(g) + H_2(g) \longrightarrow C_2H_6(g)$ $\quad \Delta H_3 = -137$ kJ mol^{-1}

Draw an enthalpy change cycle to relate the enthalpy changes of processes involved in the conversion of ethyne to ethane.

A



Practice 35.1

The oxidation of sulphur to sulphur trioxide can occur by either Route 1 or 2 as follows:

Route 1

$$S(s) + \frac{3}{2}O_2(g) \longrightarrow SO_3(g) \qquad \Delta H = -441 \text{ kJ mol}^{-1}$$

Route 2

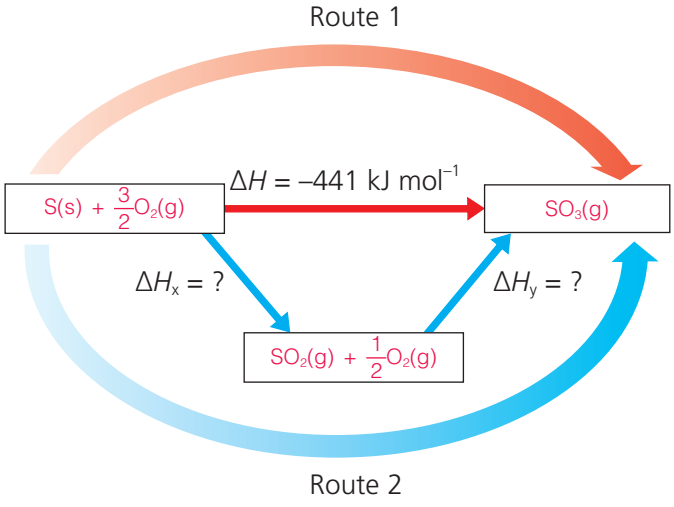
Step 1 $S(s) + O_2(g) \longrightarrow SO_2(g)$

$\Delta H = -297 \text{ kJ mol}^{-1}$

Step 2 $SO_2(g) + \frac{1}{2}O_2(g) \longrightarrow SO_3(g)$

$\Delta H = -144 \text{ kJ mol}^{-1}$

The enthalpy changes of processes involved in the oxidation of sulphur to sulphur trioxide can be related by the enthalpy change cycle shown.



a) Write down the substance(s) that should appear in each box of the enthalpy change cycle.

b) Give the values of ΔH_x and ΔH_y respectively.

$\Delta H_x = -297 \text{ kJ mol}^{-1}$ $\Delta H_y = -144 \text{ kJ mol}^{-1}$

35.2 Using Hess's Law to determine enthalpy changes that cannot be easily obtained by experiment

In the Haber process, ammonia is made from the reaction between nitrogen and hydrogen:

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g) \qquad \Delta H_r^{\ominus} = -92.2 \text{ kJ mol}^{-1}$$

If we dig into the details of the reaction, we find that it is not as simple as it looks. In fact, the overall reaction occurs by a series of steps, with hydrazine ($N_2H_4(g)$) produced at an intermediate stage:

$$N_2(g) + 2H_2(g) \longrightarrow N_2H_4(g) \xrightarrow{H_2(g)} 2NH_3(g)$$

The enthalpy change for the conversion of hydrazine to ammonia can be measured as $\Delta H^{\ominus} = -187.6 \text{ kJ mol}^{-1}$.

Suppose we are interested in the standard enthalpy change of formation of hydrazine. This corresponds to the enthalpy change of the process:

$$N_2(g) + 2H_2(g) \longrightarrow N_2H_4(g) \qquad \Delta H_f^{\ominus} = ?$$

We would have difficulty because the reaction does not go clearly. Some of the hydrazine is converted into ammonia while some of the starting nitrogen still remains.

The chemist Germain Henri Hess used the law of conservation of energy to find a way of working out the enthalpy changes that cannot be easily obtained by experiment. He developed Hess's Law, which says that the enthalpy change of a reaction depends only on the initial and final states.

✓ If a reaction can be written as the sum of several steps, the enthalpy change for the overall reaction equals the sum of enthalpy changes of the steps.

For example,

Overall reaction = *Step 1* + *Step 2* + *Step 3*

then

the enthalpy change for the overall reaction

$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$$

In the synthesis of ammonia, the sum of *Steps 1* and *2* is equal to the overall reaction.

Step 1	$N_2(g) + 2H_2(g) \longrightarrow \cancel{N_2H_4(g)}$	$\Delta H_1^{\ominus} = ?$
Step 2	$\cancel{N_2H_4(g)} + H_2(g) \longrightarrow 2NH_3(g)$	$\Delta H_2^{\ominus} = -187.6$ kJ mol^{-1}
Overall reaction	$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$	$\Delta H_r^{\ominus} = -92.2$ kJ mol^{-1}

By Hess's Law,

$$\Delta H_1^{\ominus} + \Delta H_2^{\ominus} = \Delta H_r^{\ominus}$$

then $\Delta H_1^{\ominus} = \Delta H_r^{\ominus} - \Delta H_2^{\ominus}$

$= [(-92.2) - (-187.6)]$ kJ mol^{-1}

$= +95.4$ kJ mol^{-1}

Therefore the standard enthalpy change of formation of hydrazine is +95.4 kJ mol^{-1}.

Alternatively, we can set up an enthalpy change cycle (Fig. 35.3) to relate the enthalpy changes of all the processes involved.

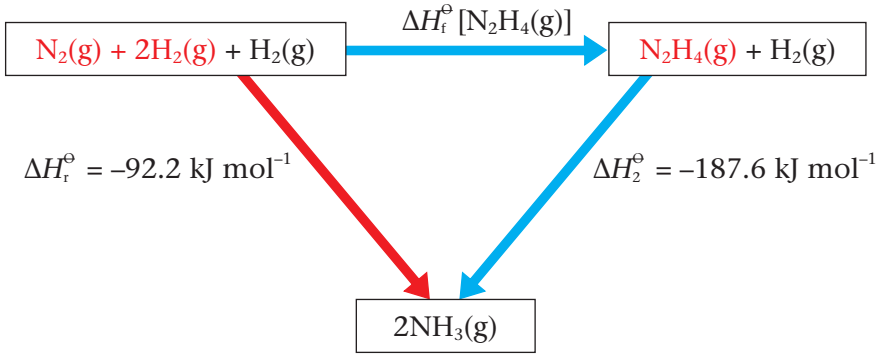


Fig. 35.3 An enthalpy change cycle for determining the standard enthalpy change of formation of hydrazine

Notice the following points for the enthalpy change cycle shown in Fig. 35.3:

- The equation of the process whose enthalpy change we try to find is written at the top of the cycle.
- Hydrogen is added to both sides of the top equation so as to balance the equations with downward pointing arrows. Adding the hydrogen does not change the process in the top equation.

By Hess's Law,

$$\Delta H_f^{\ominus}[N_2H_4(g)] = [(-92.2) - (-187.6)] \text{ kJ mol}^{-1}$$
$$= +95.4 \text{ kJ mol}^{-1}$$

Example 35.2

Q Water gas is the name for the industrially important mixture of CO(g) and $H_2(g)$ prepared by passing steam over hot charcoal at 1 000 °C:

$$C(s) + H_2O(g) \longrightarrow \underset{\text{water gas}}{CO(g) + H_2(g)}$$

Use the following information to calculate the standard enthalpy change of the water gas reaction:

Process	Chemical equation	$\Delta H^\ominus$ (kJ)
1	$C(s) + O_2(g) \longrightarrow CO_2(g)$	–394
2	$2CO(g) + O_2(g) \longrightarrow 2CO_2(g)$	–566
3	$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g)$	–484

A Look at the equation for the water gas reaction, we need 1 mole of CO(g) as the product. Equation (2) has 2 moles of CO(g), but they are on the reactant side. So, reverse the equation and multiply it by $\frac{1}{2}$, giving equation (2)'.

(2)' $CO_2(g) \longrightarrow CO(g) + \frac{1}{2}O_2(g) \quad \Delta H^\ominus = \frac{+566}{2}\text{ kJ} = +283\text{ kJ}$

If we reverse the equation of a process, change the sign of the ΔH value. If we multiply the equation of a process with a factor, multiply the ΔH value by the same factor.

Next, we need 1 mole of $H_2O(g)$ as the reactant. Equation (3) has 2 moles of $H_2O(g)$, but they are on the product side. So, reverse the equation and multiply it by $\frac{1}{2}$, giving equation (3)'.

(3)' $H_2O(g) \longrightarrow H_2(g) + \frac{1}{2}O_2(g) \quad \Delta H^\ominus = \frac{+484}{2}\text{ kJ} = +242\text{ kJ}$

Combine equations (1), (2)' and (3)', followed by collecting like terms, to obtain the equation for the water gas reaction.

(1) $C(s) + \cancel{O_2(g)} \longrightarrow \cancel{CO_2(g)} \quad \Delta H^\ominus = -394\text{ kJ}$

(2)' $\cancel{CO_2(g)} \longrightarrow CO(g) + \frac{1}{2}\cancel{O_2(g)} \quad \Delta H^\ominus = +283\text{ kJ}$

(3)' $H_2O(g) \longrightarrow H_2(g) + \frac{1}{2}\cancel{O_2(g)} \quad \Delta H^\ominus = +242\text{ kJ}$

$C(s) + H_2O(g) \longrightarrow CO(g) + H_2(g)$

By Hess's Law,

$\Delta H_r^\ominus = [(-394) + (+283) + (+242)]\text{ kJ} = +131\text{ kJ}$

$\therefore$ the standard enthalpy change of the water gas reaction is +131 kJ.

Example 35.3

Q The following experimental information is available.

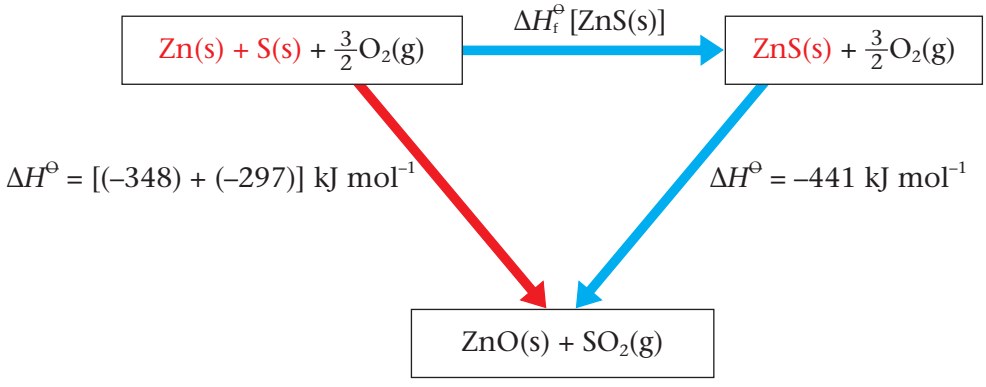
Process	Chemical equation	$\Delta H^{\ominus}$ (kJ mol^{-1})
1	$Zn(s) + \frac{1}{2}O_2(g) \longrightarrow ZnO(s)$	–348
2	$ZnS(s) + \frac{3}{2}O_2(g) \longrightarrow ZnO(s) + SO_2(g)$	–441
3	$S(s) + O_2(g) \longrightarrow SO_2(g)$	–297

The formation of zinc sulphide from its constituent elements can be represented by the following equation:

$Zn(s) + S(s) \longrightarrow ZnS(s) \qquad \Delta H_f^{\ominus}[ZnS(s)] = ?$

Using the data provided, draw an enthalpy change cycle and use it to find the standard enthalpy change of formation of ZnS(s).

A



By Hess's Law,

$\Delta H_f^{\ominus}[ZnS(s)] = [(-348) + (-297) - (-441)]$ kJ mol^{-1}
$= -204$ kJ mol^{-1}

$\therefore$ the standard enthalpy change of formation of zinc sulphide is –204 kJ mol^{-1}.

Practice 35.2

Refer to page T20.

1 The following enthalpy changes of reaction of the allotropes of sulphur are obtained from an experiment:

$S(rhombic) + O_2(g) \longrightarrow SO_2(g)$

$\Delta H = -296.0$ kJ

$S(monoclinic) + O_2(g) \longrightarrow SO_2(g)$

$\Delta H = -296.4$ kJ

a) Which allotrope of sulphur is more stable?
Rhombic sulphur is more stable.

b) Calculate the enthalpy change for the following process under the experimental conditions.
+0.4 kJ
$S(rhombic) \longrightarrow S(monoclinic)$

2 The industrial degreasing solvent dichloromethane, CH_2Cl_2, is prepared from methane by reaction with chlorine:

$CH_4(g) + 2Cl_2(g) \longrightarrow CH_2Cl_2(g) + 2HCl(g)$

The enthalpy changes of the following processes were obtained under standard conditions:

Process	Chemical equation	$\Delta H^\ominus$ (kJ)
1	$CH_4(g) + Cl_2(g) \longrightarrow CH_3Cl(g) + HCl(g)$	–98
2	$CH_3Cl(g) + Cl_2(g) \longrightarrow CH_2Cl_2(g) + HCl(g)$	–104

Draw an enthalpy change cycle to relate the enthalpy changes of all the processes involved and use it to calculate the enthalpy change of the reaction.
–202 kJ

35.3 Determining the enthalpy change of formation of magnesium hydroxide from enthalpy changes of other reactions

Video
Determining the enthalpy change of formation of magnesium oxide

35.1
Determining the enthalpy change of formation of magnesium oxide.

35.2 Assessment Form
Determining the enthalpy change of thermal decomposition of potassium hydrogencarbonate.

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Suppose we are interested in the enthalpy change of formation of magnesium hydroxide. The enthalpy change corresponds to that of the following process:

$$Mg(s) + H_2(g) + O_2(g) \longrightarrow Mg(OH)_2(s)$$

As magnesium hydroxide cannot be formed directly from its constituent elements, we cannot obtain its enthalpy change of formation directly from experiment. Instead, we can employ an indirect route, again based on Hess's Law.

Video
Determining the enthalpy change of thermal decomposition of potassium hydrogencarbonate

We can calculate the enthalpy change of formation of magnesium hydroxide from the following information:

- the enthalpy change of the reaction between magnesium and hydrochloric acid;
- the enthalpy change of the reaction between magnesium hydroxide and hydrochloric acid; and
- the enthalpy change of formation of water.

Notice that we can determine the enthalpy changes of the first two reactions experimentally and obtain the enthalpy change of formation of water from a data book.

Now we are going to see how we can determine the enthalpy changes of the mentioned reactions, and use the data to calculate the enthalpy change of formation of magnesium hydroxide.

Part A Determining the enthalpy change of the reaction between magnesium and hydrochloric acid

Allow 0.610 g of magnesium to react with excess hydrochloric acid in an expanded polystyrene cup (Fig. 35.4). Measure the maximum rise in temperature of the mixture.

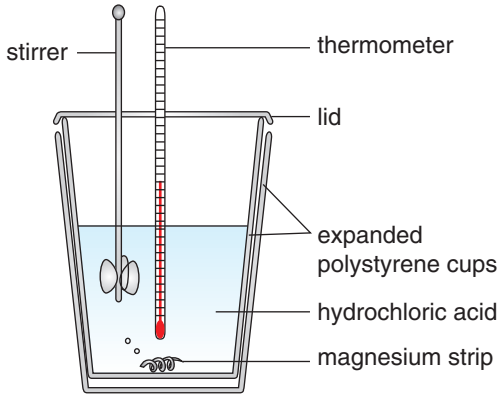


Fig. 35.4 A simple calorimeter used to determine the enthalpy change of the reaction between Mg(s) and HCl(aq)

In a certain experiment, the maximum rise in temperature of the mixture was found to be 23.8 °C.

N1 Remind students to be careful when reading a question – the question may either provide the specific heat capacity or the heat capacity of a solution.

DO NOT confuse 'specific heat capacity' with 'heat capacity'.

Assuming that the heat capacity of the acid used is 542 J K^{-1} and the expanded polystyrene cup has negligible heat capacity, we can calculate the enthalpy change of the following reaction: N1

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$$

Amount of heat released during reaction = 542 J K^{-1} x 23.8 K
= 12 900 J

$$\text{Number of moles of Mg reacted} = \frac{0.610\ \text{g}}{24.3\ \text{g mol}^{-1}}$$
$$= 0.0251\ \text{mol}$$

$$\text{Enthalpy change of the reaction} = \frac{-12\,900\ \text{J}}{0.0251\ \text{mol}}$$
$$= -514\ \text{kJ mol}^{-1}$$

Part B Determining the enthalpy change of the reaction between magnesium hydroxide and hydrochloric acid

ASL 2011 Paper 2 Q3(a)(ii) AL 2011 Paper 2 Q3(a)(ii)

Allow 4.37 g of magnesium hydroxide to react with excess hydrochloric acid in an expanded polystyrene cup (Fig. 35.5). Measure the maximum rise in temperature of the mixture.

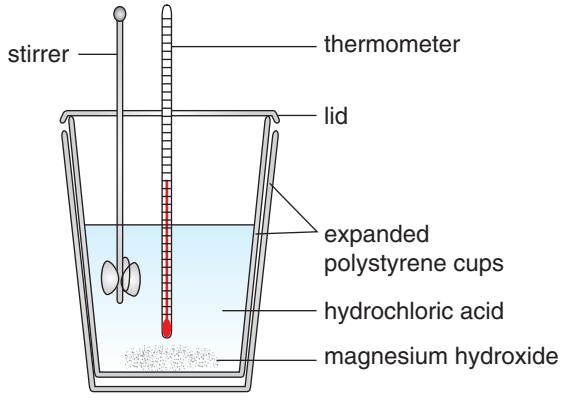


Fig. 35.5 A simple calorimeter used to determine the enthalpy change of the reaction between $Mg(OH)_2(s)$ and HCl(aq)

In a certain experiment, the maximum rise in temperature of the mixture was found to be 15.1 °C.

Assuming that the heat capacity of the acid used is 542 J K^{-1} and the expanded polystyrene cup has negligible heat capacity, we can calculate the enthalpy change for the following reaction:

$$Mg(OH)_2(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + 2H_2O(l)$$

Amount of heat released during reaction = 542 J K^{-1} x 15.1 K
= 8 180 J

Number of moles of $Mg(OH)_2$ reacted = $\frac{4.37\ g}{58.3\ g\ mol^{-1}}$
= 0.0750 mol

Enthalpy change of the reaction = $\frac{-8\ 180\ J}{0.0750\ mol}$
= –109 kJ mol^{-1}

Part C Calculating the enthalpy change of formation of magnesium hydroxide

ASL 2011 Paper 2 Q3(b) AL 2011 Paper 2 Q3(b)

$\Delta H_f[Mg(OH)_2(s)]$ refers to the enthalpy change of the following process:

$Mg(s) + H_2(g) + O_2(g) \longrightarrow Mg(OH)_2(s)$ $\Delta H_f[Mg(OH)_2(s)] = ?$

ΔH values of the following processes are known:

(1) $Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$ $\Delta H = -514$ kJ mol^{-1}

(2) $Mg(OH)_2(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + 2H_2O(l)$ $\Delta H = -109$ kJ mol^{-1}

(3) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$ $\Delta H = -286$ kJ mol^{-1}

Reverse equation (2) to give equation (2)'. Combine the equations as shown below, followed by collecting like terms, to obtain the equation for the formation of $Mg(OH)_2(s)$ from its constituent elements.

(1) $Mg(s) + \cancel{2HCl(aq)} \longrightarrow \cancel{MgCl_2(aq)} + \cancel{H_2(g)}$ $\Delta H = -514$ kJ

(2)' $\cancel{MgCl_2(aq)} + \cancel{2H_2O(l)} \longrightarrow Mg(OH)_2(s) + \cancel{2HCl(aq)}$ $\Delta H = +109$ kJ

2 x (3) $\cancel{2}H_2(g) + O_2(g) \longrightarrow \cancel{2H_2O(l)}$ $\Delta H = -572$ kJ

$Mg(s) + H_2(g) + O_2(g) \longrightarrow Mg(OH)_2(s)$

By Hess's Law,

$\Delta H_f[Mg(OH)_2(s)] = [(-514) + (+109) + (-572)]$ kJ mol^{-1}
= –977 kJ mol^{-1}

Therefore the enthalpy change of formation of magnesium hydroxide is –977 kJ mol^{-1}.

Alternatively, we can draw an enthalpy change cycle (Fig. 35.6) to relate the enthalpy changes of all the processes involved.

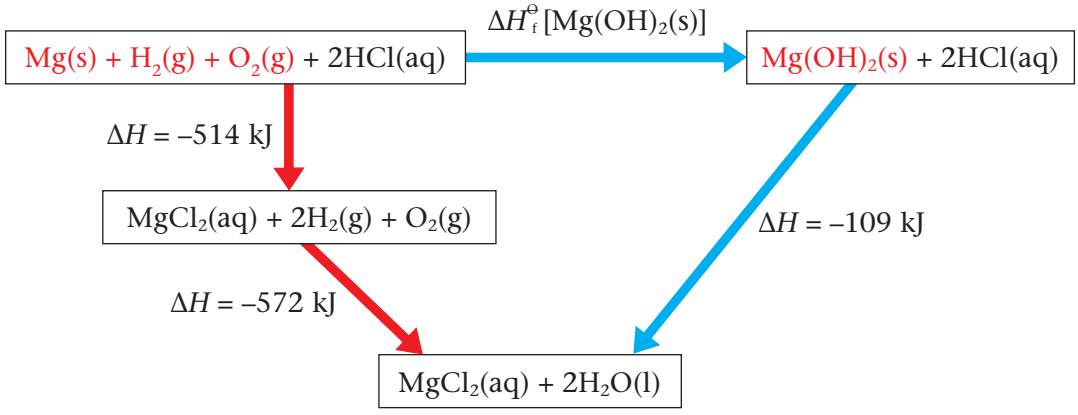


Fig. 35.6 An enthalpy change cycle for determining the standard enthalpy change of formation of magnesium hydroxide

By Hess's Law,

$$\Delta H_f^\ominus[Mg(OH)_2(s)] = [(-514) + (-572) - (-109)] \text{ kJ mol}^{-1}$$
$$= -977 \text{ kJ mol}^{-1}$$

Practice 35.3

Refer to page T20.

DSE 2012 Paper 1B Q8(c)

Upon heating, potassium hydrogencarbonate ($KHCO_3$) decomposes according to the following equation:

$2KHCO_3(s) \longrightarrow K_2CO_3(s) + H_2O(l) + CO_2(g)$

The enthalpy change of decomposition of potassium hydrogencarbonate can be determined indirectly from the enthalpy changes of the following two reactions:

1 $KHCO_3(s) + HCl(aq) \longrightarrow KCl(aq) + H_2O(l) + CO_2(g)$ $\quad \Delta H = +19.6$ kJ mol^{-1}

2 $K_2CO_3(s) + 2HCl(aq) \longrightarrow 2KCl(aq) + H_2O(l) + CO_2(g)$

In an experiment to determine the enthalpy change of *Reaction 2*, 3.45 g of $K_2CO_3(s)$ were added to excess HCl(aq) in an expanded polystyrene cup. The experimental data obtained are shown below:

Initial temperature of the reacting solution	26.2 °C
Final temperature of the reacting solution	31.4 °C
Mass of the resulting solution	54.5 g
Specific heat capacity of the resulting solution	4.3 J g^{-1} K^{-1}
Molar mass of K_2CO_3	138.0 g mol^{-1}

a) Assuming that the heat capacity of the cup used is negligible, calculate the enthalpy change of *Reaction 2* from the above data.
–48.8 kJ mol^{-1}

b) Calculate the enthalpy change of decomposition of $KHCO_3(s)$.
+44.0 kJ mol^{-1}

35.4 Determining the standard enthalpy change of formation of a compound from standard enthalpy changes of combustion

Suppose we are interested in the standard enthalpy change of formation of methane represented by the following process:

$$C(\text{graphite}) + 2H_2(g) \longrightarrow CH_4(g) \qquad \Delta H_f^{\ominus}[CH_4(g)] = ?$$

However, carbon and hydrogen do not react to produce methane under normal conditions, so we cannot measure the enthalpy change of formation of methane directly. Instead, we must employ an indirect route, again based on Hess's Law. N2

N2 The enthalpy change of the following reaction must be determined by an INDIRECT method:
$2C(\text{graphite}) + O_2(g) \longrightarrow 2CO(g)$
$CO_2(g)$ may also be formed in the process.

Since methane contains carbon and hydrogen only, we can relate them to carbon dioxide and water by the combustion reactions of methane and its constituent elements as shown in Fig. 35.7.

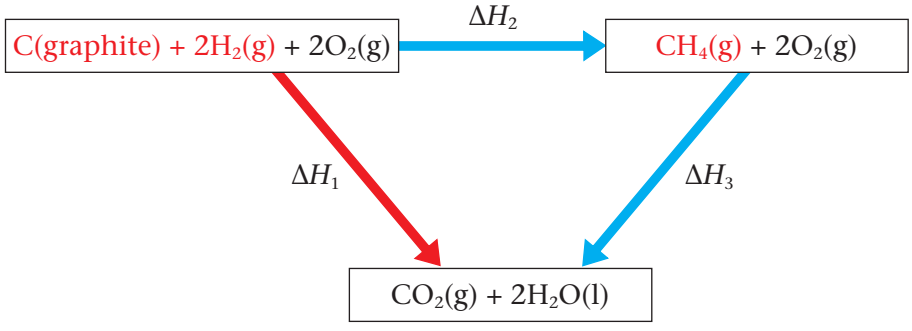


Fig. 35.7 An enthalpy change cycle relating $\Delta H_f^{\ominus}[CH_4(g)]$ to enthalpy changes of combustion

Notice the following points for the enthalpy change cycle shown in Fig. 35.7:

- The equation of the process whose enthalpy change we try to find is written at the top of the cycle.
- The combustion products, carbon dioxide and water, are written at the bottom of the cycle.
- Oxygen is added to both sides of the top equation so as to balance equations showing the combustion reactions of methane and its constituent elements (i.e. the two equations with downward pointing arrows). Adding the oxygen does not change the process in the top equation.

Refer to the enthalpy change cycle shown in Fig. 35.7.

$\Delta H_1 = \Delta H_c^{\ominus}[\text{C(graphite)}] + 2 \times \Delta H_c^{\ominus}[H_2(g)]$

$\Delta H_2 = \Delta H_f^{\ominus}[CH_4(g)]$

$\Delta H_3 = \Delta H_c^{\ominus}[CH_4(g)]$

The following $\Delta H_c^{\ominus}$ values are known:

Substance	$\Delta H_c^{\ominus}$ (kJ mol^{-1})
C(graphite)	–394
$H_2(g)$	–286
$CH_4(g)$	–890

By Hess's Law, we can write

$\Delta H_1 = \Delta H_2 + \Delta H_3$

Rearranging the equation, we have $\Delta H_2 = \Delta H_1 - \Delta H_3$

So, $\Delta H_2 = \Delta H_f^{\ominus}[CH_4(g)]$
$= \Delta H_1 - \Delta H_3$
$= \Delta H_c^{\ominus}[\text{C(graphite)}] + 2 \times \Delta H_c^{\ominus}[H_2(g)] - \Delta H_c^{\ominus}[CH_4(g)]$
$= [(-394) + 2(-286) - (-890)]$ kJ mol^{-1}
$= -76$ kJ mol^{-1}

From the above discussion, we know that

$\Delta H_f^{\ominus}[CH_4(g)] = \Delta H_c^{\ominus}[\text{C(graphite)}] + 2 \times \Delta H_c^{\ominus}[H_2(g)] - \Delta H_c^{\ominus}[CH_4(g)]$

When a compound is completely burnt, the combustion products produced are the same as those produced when its constituent elements undergo complete combustion. Hence the standard enthalpy change of formation of a compound is related to the standard enthalpy changes of combustion of both the compound and its constituent elements.

By Hess's Law,

Standard enthalpy change of formation of a compound	=	Sum of standard enthalpy changes of combustion of the constituent elements	–	Standard enthalpy change of combustion of the compound

✓ $\Delta H_f^\ominus$[compound]
= $\Sigma\ \Delta H_c^\ominus$[constituent elements] – $\Delta H_c^\ominus$[compound]

Fig. 35.8 shows a general enthalpy change cycle relating the standard enthalpy change of formation of a compound to standard enthalpy changes of combustion.

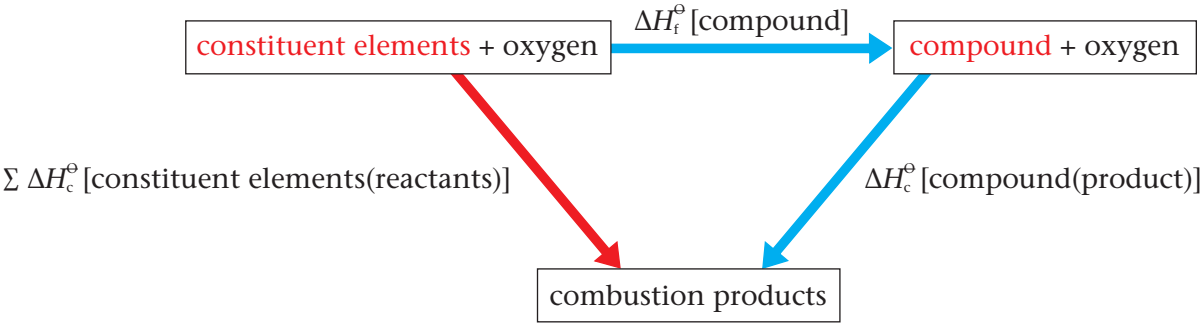


Fig. 35.8 A general enthalpy change cycle relating $\Delta H_f^\ominus$ of a compound to standard enthalpy changes of combustion

Example 35.4

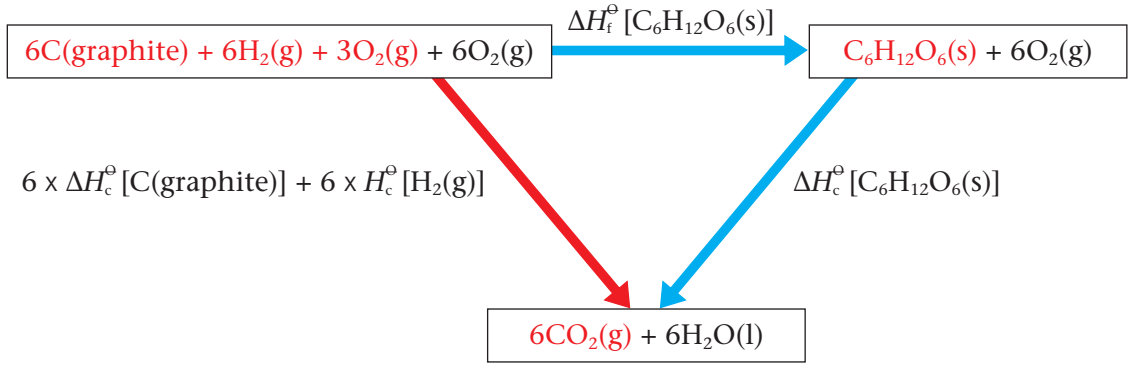
Q Using the data below, draw an enthalpy change cycle and use it to calculate the standard enthalpy change of formation of glucose ($C_6H_{12}O_6(s)$).

Substance	$\Delta H_c^\ominus$ (kJ mol^{-1})
C(graphite)	–394
$H_2(g)$	–286
$C_6H_{12}O_6(s)$	–2 803

Continued on next page

A $\Delta H_f^{\ominus}[C_6H_{12}O_6(s)]$ refers to the standard enthalpy change of the following process:

$6C(\text{graphite}) + 6H_2(g) + 3O_2(g) \longrightarrow C_6H_{12}O_6(s)$



By Hess's Law

$\Delta H_f^{\ominus}[C_6H_{12}O_6(s)] = [6(-394) + 6(-286) - (-2\,803)]\ \text{kJ mol}^{-1}$
$= -1\,277\ \text{kJ mol}^{-1}$

$\therefore$ the standard enthalpy change of formation of glucose is $-1\,277\ \text{kJ mol}^{-1}$.

Example 35.5

Q Using the data below, calculate the standard enthalpy change of formation of ethanol.

Substance	$\Delta H_c^{\ominus}$ (kJ mol^{-1})
C(graphite)	–394
$H_2(g)$	–286
$C_2H_5OH(l)$	–1 368

DSE 2012 Paper 1A Q7

A $\Delta H_f^{\ominus}[C_2H_5OH(l)]$ refers to the standard enthalpy change of the following process:

$2C(\text{graphite}) + 3H_2(g) + \frac{1}{2}O_2(g) \longrightarrow C_2H_5OH(l)$

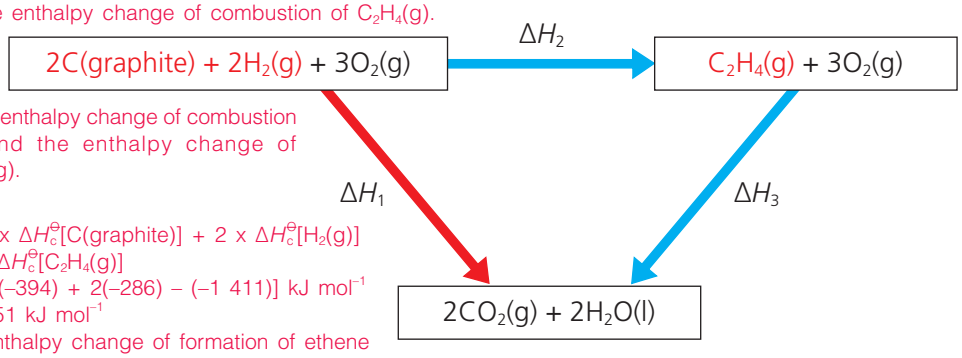
$\Delta H_f^{\ominus}[\text{compound}] = \Sigma\ \Delta H_c^{\ominus}[\text{constituent elements}] - \Delta H_c^{\ominus}[\text{compound}]$

$\Delta H_f^{\ominus}[C_2H_5OH(l)] = 2 \times \Delta H_c^{\ominus}[C(\text{graphite})] + 3 \times \Delta H_c^{\ominus}[H_2(g)] - \Delta H_c^{\ominus}[C_2H_5OH(l)]$
$= [2(-394) + 3(-286) - (-1\,368)]\ \text{kJ mol}^{-1}$
$= -278\ \text{kJ mol}^{-1}$

$\therefore$ the standard enthalpy change of formation of ethanol is $-278\ \text{kJ mol}^{-1}$.

Practice 35.4

1 Study the following enthalpy change cycle which is used to find the enthalpy change of formation of ethene.



a) ΔH_2 represents the enthalpy change of formation of $C_2H_4(g)$.
ΔH_3 represents the enthalpy change of combustion of $C_2H_4(g)$.

b) ΔH_1 represents the enthalpy change of combustion of C(graphite) and the enthalpy change of combustion of $H_2(g)$.

c) $\Delta H_f^{\ominus}[C_2H_4(g)] = 2 \times \Delta H_c^{\ominus}[C(\text{graphite})] + 2 \times \Delta H_c^{\ominus}[H_2(g)] - \Delta H_c^{\ominus}[C_2H_4(g)]$
$= [2(-394) + 2(-286) - (-1\,411)]$ kJ mol^{-1}
$= +51$ kJ mol^{-1}
$\therefore$ the standard enthalpy change of formation of ethene is +51 kJ mol^{-1}.

a) What are the enthalpy changes represented by ΔH_2 and ΔH_3 respectively?

b) What are the combined enthalpy changes represented by ΔH_1?

c) The following standard enthalpy changes of combustion are obtained from experiments:

Substance	$\Delta H_c^{\ominus}$ (kJ mol^{-1})
C(graphite)	–394
$H_2(g)$	–286
$C_2H_4(g)$	–1 411

Calculate the standard enthalpy change of formation of ethene.

d) Suggest a reason why it is difficult to determine the standard enthalpy change of formation of ethene directly from experiments.
Carbon and hydrogen do not react to produce ethene under normal conditions.

2 Using the data given below, calculate the standard enthalpy change of formation of sucrose ($C_{12}H_{22}O_{11}$).

Substance	$\Delta H_c^{\ominus}$ (kJ mol^{-1})
C(graphite)	–394
$H_2(g)$	–286
$C_{12}H_{22}O_{11}(s)$	–5 640

Practice 35.4

2 $\Delta H_f^{\ominus}[C_{12}H_{22}O_{11}(s)]$ refers to the standard enthalpy change of the following process:

$$12C(\text{graphite}) + 11H_2(g) + \frac{11}{2}O_2(g) \longrightarrow C_{12}H_{22}O_{11}(s)$$

$\Delta H_f^{\ominus}[\text{compound}] = \sum \Delta H_c^{\ominus}[\text{constituent elements}] - \Delta H_c^{\ominus}[\text{compound}]$
$\Delta H_f^{\ominus}[C_{12}H_{22}O_{11}(s)] = 12 \times \Delta H_c^{\ominus}[C(\text{graphite})] + 11 \times \Delta H_c^{\ominus}[H_2(g)] - \Delta H_c^{\ominus}[C_{12}H_{22}O_{11}(s)]$
$= [12(-394) + 11(-286) - (-5\,640)]$ kJ mol^{-1}
$= -2\,230$ kJ mol^{-1}
$\therefore$ the standard enthalpy change of formation of sucrose is –2 230 kJ mol^{-1}.

35.5 Determining the standard enthalpy change of a reaction from standard enthalpy changes of formation

Many data books provide detailed tables of enthalpy changes of formation of compounds. These data can be used to calculate the enthalpy changes of reactions, without having to do experiments to measure them.

Chemists need information about enthalpy changes of reactions when designing a new chemical production plant. In such a plant, an exothermic process may release sufficient energy to cause a fire or explosion. Careful design allows the release of energy to be controlled and even re-used.

Imagine that you are building a plant to make slaked lime ($Ca(OH)_2$) from quicklime (CaO). The equation for this exothermic reaction is:

$$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(s) \qquad \Delta H_r^{\ominus} = ?$$

The standard enthalpy changes of formation of the reactants and the product are known:

Substance	$\Delta H_f^{\ominus}$ (kJ mol^{-1})
CaO(s)	–635
$H_2O(l)$	–286
$Ca(OH)_2(s)$	–986

We can relate the enthalpy changes of the processes involved using the enthalpy change cycle shown in Fig. 35.9.

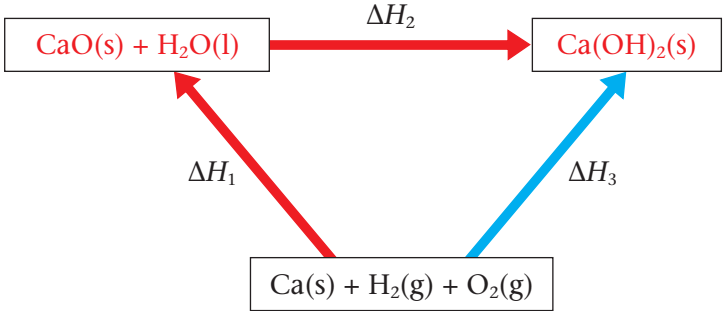


Fig. 35.9 An enthalpy change cycle relating the standard enthalpy change of the reaction between calcium oxide and water to standard enthalpy changes of formation

Notice the following points for the enthalpy change cycle shown in Fig. 35.9:

- The equation of the process whose enthalpy change we try to find is written at the top of the cycle.
- The constituent elements of $Ca(OH)_2(s)$ are written at the bottom of the cycle.
- As both $CaO(s)$ and $H_2O(l)$ appear at the top left-hand corner of the cycle, we must add the enthalpy changes of formation of the compounds when calculating ΔH_1.

Refer to the enthalpy change cycle shown in Fig. 35.9.

$$\Delta H_1 = \Delta H_f^{\ominus}[CaO(s)] + \Delta H_f^{\ominus}[H_2O(l)]$$

$$\Delta H_2 = \Delta H_r^{\ominus}$$

$$\Delta H_3 = \Delta H_f^{\ominus}[Ca(OH)_2(s)]$$

By Hess's Law, we can write

$$\Delta H_1 + \Delta H_2 = \Delta H_3$$

Rearranging the equation, we have

$$\Delta H_2 = \Delta H_3 - \Delta H_1$$

So, $\Delta H_2 = \Delta H_r^{\ominus}$

$= \Delta H_3 - \Delta H_1$

$= \Delta H_f^{\ominus}[Ca(OH)_2(s)] - (\Delta H_f^{\ominus}[CaO(s)] + \Delta H_f^{\ominus}[H_2O(l)])$

$= -986 \text{ kJ mol}^{-1} - [(-635) + (-286)] \text{ kJ mol}^{-1}$

$= -65 \text{ kJ mol}^{-1}$

We can find the standard enthalpy change of the reaction

$$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(s)$$

by using the following equation:

$$\Delta H_r^{\ominus} = \Delta H_f^{\ominus}[Ca(OH)_2(s)] - (\Delta H_f^{\ominus}[CaO(s)] + \Delta H_f^{\ominus}[H_2O(l)])$$

Thus, the standard enthalpy change of the reaction can be obtained from the standard enthalpy changes of formation of the product ($Ca(OH)_2(s)$) and the reactants ($CaO(s)$ and $H_2O(l)$).

We can find the standard enthalpy changes of other reactions in a similar manner.

Standard enthalpy change of a reaction	=	Sum of standard enthalpy changes of formation of products	−	Sum of standard enthalpy changes of formation of reactants

✓ $\Delta H_r^\ominus = \Sigma\ \Delta H_f^\ominus[\text{products}] - \Sigma\ \Delta H_f^\ominus[\text{reactants}]$

DSE 2012 Paper 1B Q8(d)(i) DSE 2014 Paper 1B Q6(b)(ii) AL 2010 Paper 2 Q2(a)(ii)

Fig. 35.10 shows a general enthalpy change cycle relating the standard enthalpy change of a reaction to standard enthalpy changes of formation.

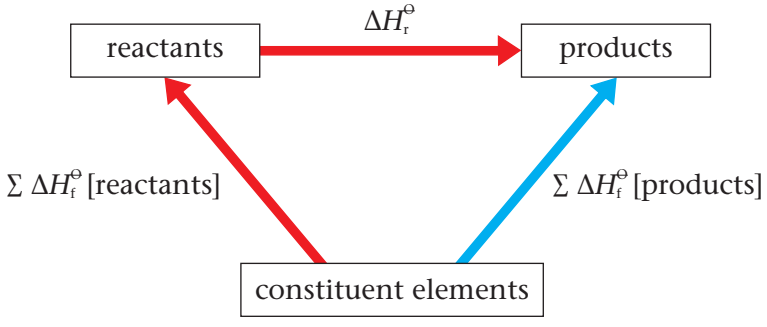


Fig. 35.10 A general enthalpy change cycle relating the standard enthalpy change of a reaction to standard enthalpy changes of formation

Example 35.6

ASL 2010 Paper 2 Q7(b)(ii)

Q Hydrazine (N_2H_4) is a liquid commonly used as a rocket fuel. It can be synthesized in a chemical process in which ammonia is oxidized by hydrogen peroxide:

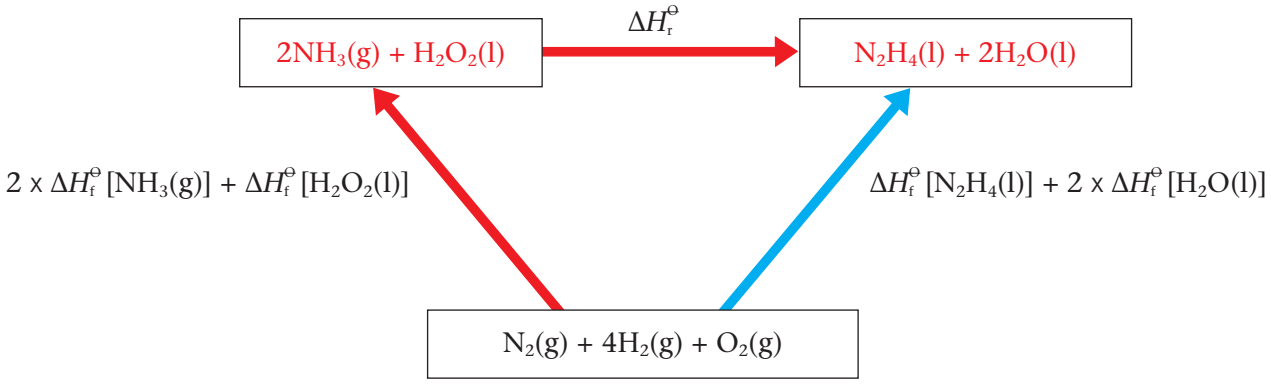
$$2NH_3(g) + H_2O_2(l) \longrightarrow N_2H_4(l) + 2H_2O(l)$$

Using the data below, draw an enthalpy change cycle and use it to calculate the standard enthalpy change of the above reaction.

Compound	$\Delta H_f^\ominus$ (kJ mol^{-1})
$H_2O(l)$	−286
$H_2O_2(l)$	−188
$NH_3(g)$	−46
$N_2H_4(l)$	+51

Continued on next page

A



$\Delta H_r^\ominus = \Delta H_f^\ominus[N_2H_4(g)] + 2 \times \Delta H_f^\ominus[H_2O(l)] - 2 \times \Delta H_f^\ominus[NH_3(g)] - \Delta H_f^\ominus[H_2O_2(l)]$

$= [(+51) + 2(-286) - 2(-46) - (-188)]$ kJ

$= -241$ kJ

$\therefore$ the standard enthalpy change of the reaction is –241 kJ.

Example 35.7

AL 2008 Paper 2 Q1(c)(ii)(II)

Q $SiCl_4(l)$ undergoes hydrolysis according to the following equation:

$$SiCl_4(l) + 2H_2O(l) \longrightarrow SiO_2(s) + 4HCl(g)$$

Using the $\Delta H_f^\ominus$ values given below, calculate the standard enthalpy change of the hydrolysis.

Substance	$\Delta H_f^\ominus$ (kJ mol^{-1})
$H_2O(l)$	–286
$HCl(g)$	–92
$SiO_2(s)$	–910
$SiCl_4(l)$	–640

A $\Delta H_r^\ominus = \Sigma \Delta H_f^\ominus[\text{products}] - \Sigma \Delta H_f^\ominus[\text{reactants}]$

$= \Delta H_f^\ominus[SiO_2(s)] + 4 \times \Delta H_f^\ominus[HCl(g)] - \Delta H_f^\ominus[SiCl_4(l)] - 2 \times \Delta H_f^\ominus[H_2O(l)]$

$= [(-910) + 4(-92) - (-640) - 2(-286)]$ kJ

$= -66$ kJ

$\therefore$ the standard enthalpy change of the hydrolysis is –66 kJ.

Example 35.8

Q Under standard conditions, complete combustion of 0.250 mole of ethane (C_2H_6) gives 390 kJ of heat. The standard enthalpy changes of formation of $H_2O(l)$ and $CO_2(g)$ are –286 kJ mol^{-1} and –394 kJ mol^{-1} respectively. What is the standard enthalpy change of formation of ethane?

DSE 2013 Paper 1A Q18
ASL 2006 Q11(a)(ii) & (iii)

A Standard enthalpy change of combustion of ethane $= \dfrac{-390 \text{ kJ}}{0.250 \text{ mol}} = -1\,560 \text{ kJ mol}^{-1}$

The standard enthalpy change of combustion of ethane refers to the standard enthalpy change of the following process:

$$C_2H_6(g) + \frac{7}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l) \qquad \Delta H_c^{\ominus}[C_2H_6(g)] = -1\,560 \text{ kJ mol}^{-1}$$

$$\Delta H_r^{\ominus} = \Sigma\, \Delta H_f^{\ominus}[\text{products}] - \Sigma\, \Delta H_f^{\ominus}[\text{reactants}]$$
$$= 2 \times \Delta H_f^{\ominus}[CO_2(g)] + 3 \times \Delta H_f^{\ominus}[H_2O(l)] - \Delta H_f^{\ominus}[C_2H_6(g)]$$

i.e. $\Delta H_f^{\ominus}[C_2H_6(g)] = 2 \times \Delta H_f^{\ominus}[CO_2(g)] + 3 \times \Delta H_f^{\ominus}[H_2O(l)] - \Delta H_r^{\ominus}$
$= [2(-394) + 3(-286) - (-1\,560)] \text{ kJ mol}^{-1}$
$= -86 \text{ kJ mol}^{-1}$

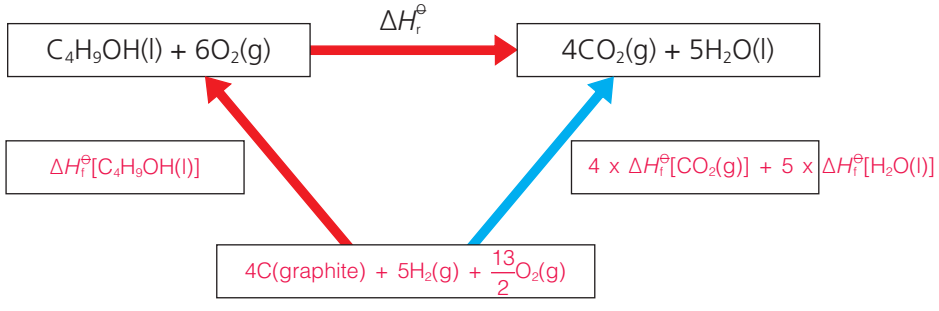
$\therefore$ the standard enthalpy change of formation of ethane is –86 kJ mol^{-1}.

Practice 35.5

1 Butan-1-ol burns in oxygen to form carbon dioxide and water as shown in the following equation:

$C_4H_9OH(l) + 6O_2(g) \longrightarrow 4CO_2(g) + 5H_2O(l)$

a) Complete the enthalpy change cycle shown below, so that you can calculate the standard enthalpy change of the above reaction ($\Delta H_r^{\ominus}$) from standard enthalpy changes of formation.



AL 2005 Paper 2 Q5(a)(i)

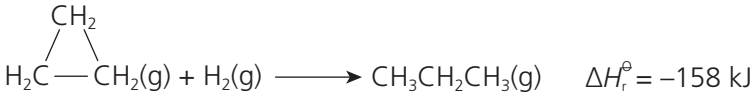
b) Calculate $\Delta H_r^{\ominus}$ of the reaction using the following data:

Refer to next page.

Substance	$\Delta H_f^{\ominus}$ (kJ mol^{-1})
$C_4H_9OH(l)$	–327
$H_2O(l)$	–286
$CO_2(g)$	–394

Continued on next page

2 Hydrogenation for cyclopropane gives propane.



The standard enthalpy changes of formation of propane is –105 kJ mol^{-1}. Calculate the standard enthalpy change of formation of cyclopropane.

Practice 35.5

1 b) By Hess's Law, we can write

$\Delta H_f^\ominus[C_4H_9OH(l)] + \Delta H_r^\ominus = 4 \times \Delta H_f^\ominus[CO_2(g)] + 5 \times \Delta H_f^\ominus[H_2O(l)]$

Rearranging the equation, we have

$\Delta H_f^\ominus = 4 \times \Delta H_f^\ominus[CO_2(g)] + 5 \times \Delta H_f^\ominus[H_2O(l)] - \Delta H_f^\ominus[C_4H_9OH(l)]$

$= [4(-394) + 5(-286) - (-327)]$ kJ

$= -2\,680$ kJ

$\therefore$ the standard enthalpy change of the reaction is –2 680 kJ.

2 $\Delta H_r^\ominus = \Delta H_f^\ominus[\text{product}] - \Sigma\Delta H_f^\ominus[\text{reactants}]$

$= \Delta H_f^\ominus[CH_3CH_2CH_3(g)] - \Delta H_f^\ominus[\text{cyclopropane}]$

$\Delta H_f^\ominus[\text{cyclopropane}] = \Delta H_f^\ominus[CH_3CH_2CH_3(g)] - \Delta H_r^\ominus$

$= [(-105) - (-158)]$ kJ mol^{-1}

$= +53$ kJ mol^{-1}

Have you mastered?

Key terms

Hess's Law 赫斯定律 49 enthalpy change cycle 焓變循環 50

Checklist

After studying this unit, you should be able to

- ☐ apply Hess's Law to construct simple enthalpy change cycles;
- ☐ apply Hess's Law to determine enthalpy changes which cannot be easily determined by experiments directly, e.g. enthalpy changes of formation of metal hydroxides;
- ☐ calculate the standard enthalpy change of formation from standard enthalpy changes of combustion;
- ☐ calculate the standard enthalpy change of reaction from standard enthalpy changes of formation.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Hess's Law states:

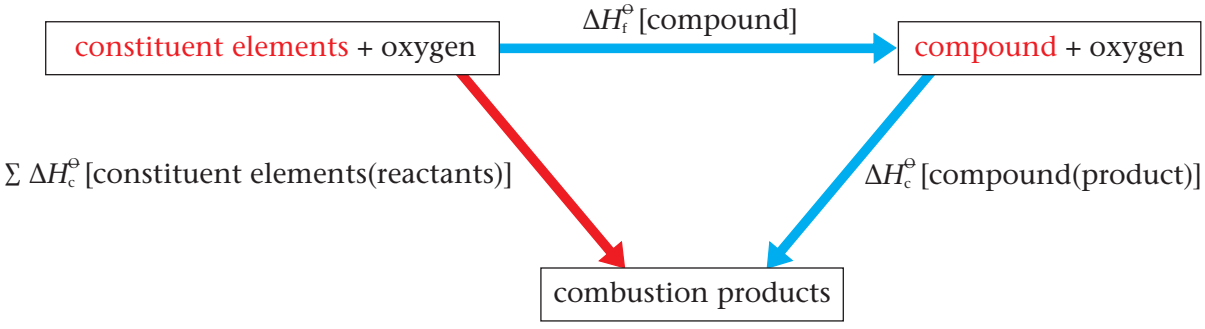
The enthalpy change of a reaction depends on the initial and final states of the reaction and is independent of the route by which the reaction may occur.

2 Overall reaction = *Step 1* + *Step 2* + *Step 3*, then

the enthalpy change for the overall reaction $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$

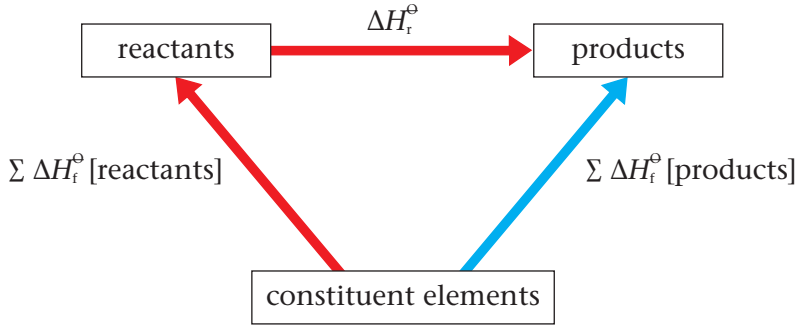
This concept enables us to find enthalpy changes that cannot be easily obtained by experiments.

3 A general enthalpy change cycle for finding the standard enthalpy change of formation of a compound from standard enthalpy changes of combustion is shown below:



$$\Delta H_f^\ominus[\text{compound}] = \Sigma\ \Delta H_c^\ominus[\text{constituent elements}] - \Delta H_c^\ominus[\text{compound}]$$

4 A general enthalpy change cycle for finding the standard enthalpy change of a reaction from standard enthalpy changes of formation is shown below:



$$\Delta H_r^\ominus = \Sigma\ \Delta H_f^\ominus[\text{products}] - \Sigma\ \Delta H_f^\ominus[\text{reactants}]$$

Have you mastered?

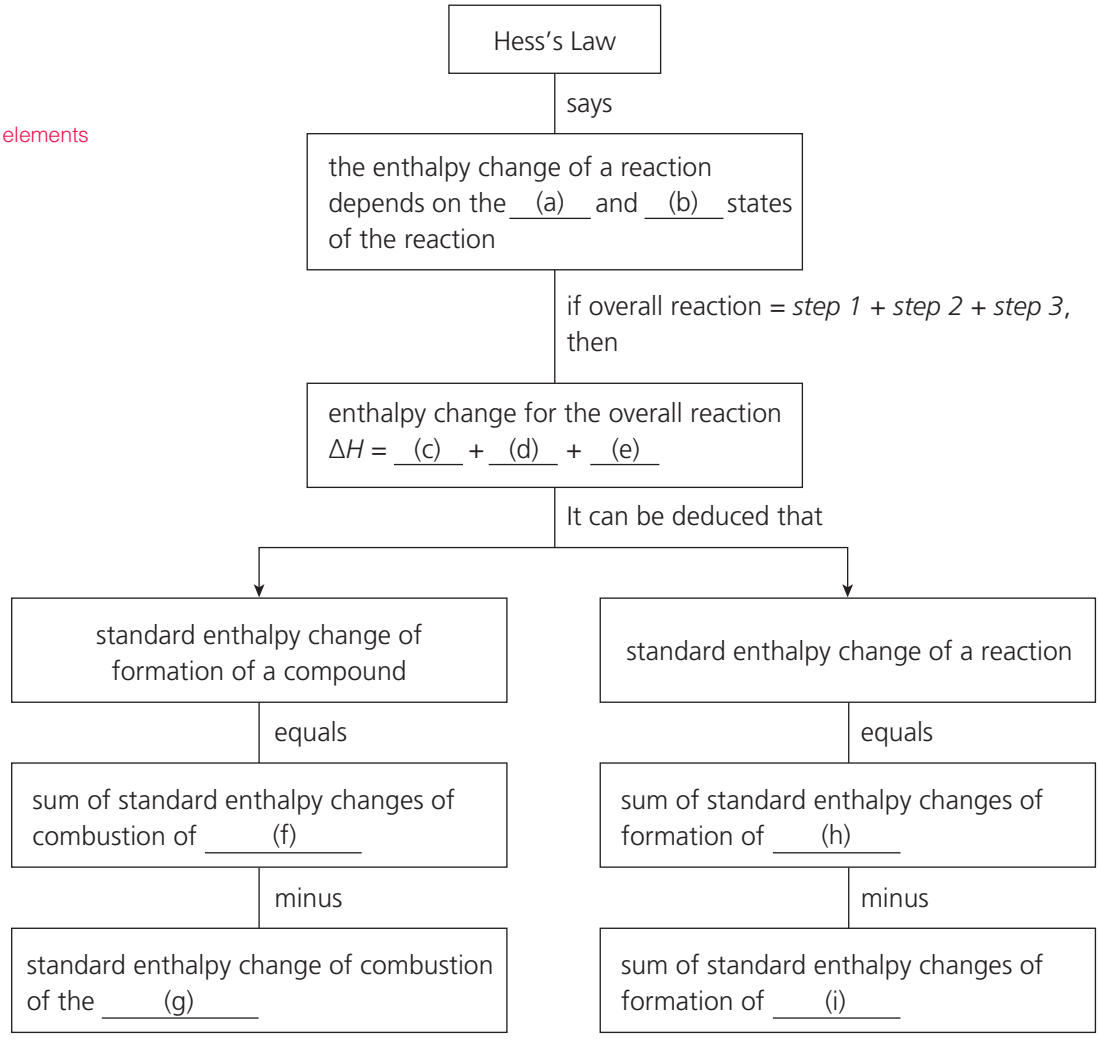
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.

a) initial
b) final
c) ΔH_1
d) ΔH_2
e) ΔH_3
f) constituent elements
g) compound
h) products
i) reactants

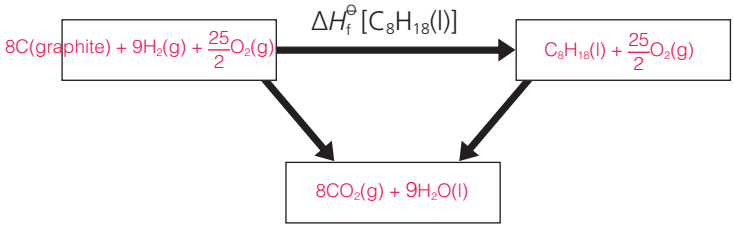


2 Octane (C_8H_{18}) is a component of petrol.

a) Write the equation which corresponds to the enthalpy change of combustion of octane.

$C_8H_{18}(l) + \frac{25}{2}O_2(g) \longrightarrow 8CO_2(g) + 9H_2O(l)$

b) Complete the enthalpy change cycle below so that you can calculate the standard enthalpy change of formation of octane, $\Delta H_f^{\ominus}[C_8H_{18}(l)]$, from standard enthalpy changes of combustion.



c) Calculate $\Delta H_f^{\ominus}[C_8H_{18}(l)]$ using the following data.

Substance	$\Delta H_c^{\ominus}$ (kJ mol^{-1})
$C_8H_{18}(l)$	–5 470
$CO_2(g)$	–394
$H_2O(l)$	–286

$\Delta H_f^{\ominus}[\text{compound}] = \sum \Delta H_c^{\ominus}[\text{constituent elements}] - \Delta H_c^{\ominus}[\text{compound}]$

$\Delta H_f^{\ominus}[C_8H_{18}(l)] = 8 \times \Delta H_c^{\ominus}[C(\text{graphite})] + 9 \times \Delta H_c^{\ominus}[H_2(g)] - \Delta H_c^{\ominus}[C_8H_{18}(l)]$

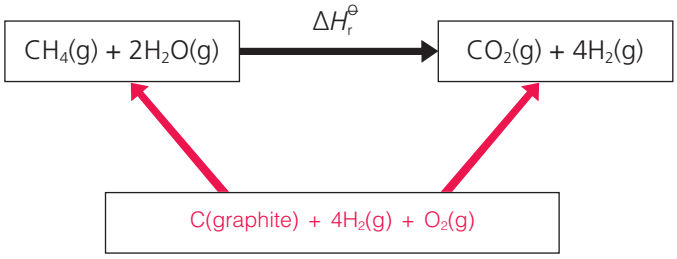
$= [8(-394) + 9(-286) - (-5\ 470)]$ kJ mol^{-1}

$= -256$ kJ mol^{-1}

3 Methane is the feedstock in the manufacture of hydrogen according to the equation:

$CH_4(g) + 2H_2O(g) \longrightarrow CO_2(g) + 4H_2(g)$

a) Complete the enthalpy change cycle for the reaction so that you can calculate the standard enthalpy change of the reaction, $\Delta H_r^{\ominus}$, from standard enthalpy changes of formation.



b) What is the value of $\Delta H_f^{\ominus}[H_2(g)]$?

$\Delta H_f^{\ominus}[H_2(g)] = 0$

c) Calculate $\Delta H_r^{\ominus}$ of the reaction using the following data.

Substance	$\Delta H_f^{\ominus}$ (kJ mol^{-1})
$CH_4(g)$	–74.8
$CO_2(g)$	–394
$H_2O(g)$	–242

$\Delta H_r^{\ominus} = \sum \Delta H_f^{\ominus}[\text{products}] - \sum \Delta H_f^{\ominus}[\text{reactants}]$

$\Delta H_r^{\ominus} = \Delta H_f^{\ominus}[CO_2(g)] - \Delta H_f^{\ominus}[CH_4(g)] - 2 \times \Delta H_f^{\ominus}[H_2O(g)]$

$= [(-394) - (-74.8) - 2(-242)]$ kJ mol^{-1}

$= +165$ kJ mol^{-1}

Unit Exercise

Part II Multiple choice questions

4 The standard enthalpy changes of combustion of some substances are shown below:

Substance	Standard enthalpy change of combustion at 298 K (kJ mol^{-1})
$H_2(g)$	–286
C(graphite)	–394
$CH_3CH_2OH(l)$	–1 371

The standard enthalpy change of formation at 298 K of $CH_3CH_2OH(l)$ is

(A) –275 kJ mol^{-1}.
B +275 kJ mol^{-1}.
C +691 kJ mol^{-1}.
D –3 017 kJ mol^{-1}.

(HKDSE, Paper 1A, 2012, 7)

5 Consider the enthalpy changes of the following reactions:

$N_2(g) + 2O_2(g) \longrightarrow 2NO_2(g)$ $\quad \Delta H = +68$ kJ

$N_2(g) + 2O_2(g) \longrightarrow N_2O_4(g)$ $\quad \Delta H = +10$ kJ

What is the enthalpy change of the reaction

$NO_2(g) \longrightarrow \frac{1}{2}N_2O_4(g)$?

Refer to page T21 for explanations.

A +58 kJ
B +29 kJ
(C) –29 kJ
D –58 kJ

6 For which of the following reactions must its enthalpy change be determined by INDIRECT methods?

A $Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$
B $2C(s) + O_2(g) \longrightarrow 2CO(g)$
C $CH_3CH_2OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$
D $MgO(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2O(l)$

(HKDSE, Paper 1A, 2013, 15)

7 The standard enthalpy changes of combustion of carbon, hydrogen and methane are shown in the table below.

Substance	$\Delta H_c^\ominus$ (kJ mol^{-1})
Carbon, C(graphite)	–394
Hydrogen, $H_2(g)$	–286
Methane, $CH_4(g)$	–891

Which one of following expressions gives the correct value for the standard enthalpy change of formation of methane in kJ mol^{-1}?

$C(graphite) + 2H_2(g) \longrightarrow CH_4(g)$

A 394 + (2 x 286) – 891
B –394 – (2 x 286) + 891
C 394 + 286 – 891
D –394 – 286 + 891

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2009, 5)

8 Given the following data:

$\Delta H_f^\ominus[FeO(s)] = -270$ kJ mol^{-1}

$\Delta H_f^\ominus[Fe_2O_3(s)] = -820$ kJ mol^{-1}

Select the expression which gives the enthalpy change, in kJ, of the reaction:

$$2FeO(s) + \frac{1}{2}O_2(g) \longrightarrow Fe_2O_3(s)$$

Refer to page T22 for explanations.

A $(-820 \times \frac{1}{2}) + 270 = -140$

B $(+820 \times \frac{1}{2}) - 270 = +140$

C –820 + (270 x 2) = –280
D +820 – (270 x 2) = +280

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2009, 7)

9 The following $\Delta H_f^\ominus$ values are known:

Substance	$\Delta H_f^\ominus$ (kJ mol^{-1})
$C_2H_6(g)$	–84.7
CO(g)	–111
$H_2O(l)$	–286

What is the standard enthalpy change of the following reaction?

$$C_2H_6(g) + \frac{5}{2}O_2(g) \longrightarrow 2CO(g) + 3H_2O(l)$$

Refer to page T22 for explanations.

A +1 170 kJ
B +995 kJ
C –995 kJ
D –1 170 kJ

10 The standard enthalpy change of combustion of propanone (CH_3COCH_3) is –1 786 kJ mol^{-1}. Which of the following is the standard enthalpy change of formation of propanone?

(Standard enthalpy changes of formation of $H_2O(l)$ and $CO_2(g)$ are –286 kJ mol^{-1} and –394 kJ mol^{-1} respectively.)

Refer to page T22 for explanations.

A +254 kJ mol^{-1}
B –254 kJ mol^{-1}
C +1 462 kJ mol^{-1}
D –1 462 kJ mol^{-1}

11 The enthalpy change of the reaction between calcium and water can be measured in the laboratory.

$Ca(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(s) + H_2(g)$

What information, other than that obtained in this experiment, is needed to calculate the enthalpy change of formation of $Ca(OH)_2(s)$?

Refer to page T22 for explanations.

A Enthalpy change of formation of hydrogen
B Enthalpy change of combustion of hydrogen
C Enthalpy change of formation of calcium
D Enthalpy change of combustion of calcium

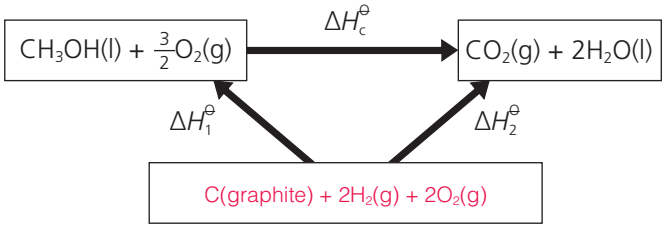
7 $\Delta H_f^\ominus[compound] = \sum \Delta H_c^\ominus[constituent\ elements] - H_c^\ominus[compound]$
$\Delta H_c^\ominus[CH_4(g)] = \Delta H_c^\ominus[C(graphite)] + 2 \times \Delta H_c^\ominus[H_2(g)] - \Delta H_c^\ominus[CH_4(g)]$
$= [(-394) + 2(-286) - (-891)]$ kJ mol^{-1}
$= [-394 - (2 \times 286) + 891]$ kJ mol^{-1}

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

12 a) The enthalpy change cycle below can be used to calculate the standard enthalpy change of combustion of methanol, using standard enthalpy changes of formation.

Refer to page T22.



i) Complete the cycle by filling in the empty box.

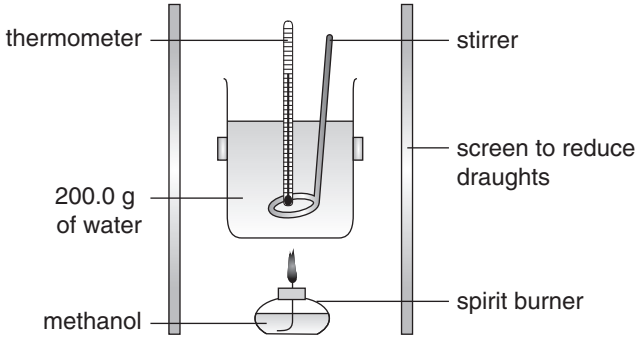
ii) Define the term 'standard enthalpy change of formation' of a compound, making clear the meaning of 'standard' in this context.

iii) Use your cycle and the data below to calculate the standard enthalpy change of combustion of methanol, $\Delta H_c^\ominus$.

–727 kJ mol^{-1}

Substance	$\Delta H_f^\ominus$ (kJ mol^{-1})
$CO_2(g)$	–394
$H_2O(l)$	–286
$CH_3OH(l)$	–239

b) An experiment was carried out, using the apparatus below, to estimate the enthalpy change of combustion of methanol.



After burning the methanol for a few minutes, the temperature of water had risen by 20.7 °C and the mass of methanol burnt was 0.848 g.

i) Calculate the amount of heat transferred to the water.

17 300 J

Heat transferred (J)
= mass of water x 4.18 x temperature change

ii) Calculate the number of moles of methanol, CH_3OH, burnt during the experiment.

0.0265 mol

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

iii) Use your answers to (b)(i) and (ii) to calculate the experimental value for the enthalpy change of combustion.

–653 kJ mol^{-1}

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2010, 17(b)-(c))

13 The table below lists the standard enthalpy changes of combustion of three substances.

Refer to page T23.

Substance	$\Delta H_c^\ominus$ (kJ mol^{-1})
C(graphite)	–394
$H_2(g)$	–286
$CH_3C{\equiv}CH(g)$ Propyne	–1 938

a) Write the equation which corresponds to the enthalpy change of formation of propyne ($CH_3C{\equiv}CH$).

3C(graphite) + $2H_2(g)$ ⟶ $CH_3C{\equiv}CH(g)$

b) Use the above enthalpy changes to construct an enthalpy change cycle and use it to calculate the standard enthalpy change of formation of propyne.

+184 kJ mol^{-1}

c) Do you agree with the following statement? Explain your answer.

'The standard enthalpy change of formation of propyne can be determined directly by experiment.'

Unit Exercise

14 a) On strong heating, calcium carbonate decomposes to calcium oxide and carbon dioxide:

$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$

Owing to the conditions under which the reaction occurs, it is not possible to measure the enthalpy change directly.

An indirect method employs the enthalpy changes when calcium carbonate and calcium oxide react with hydrochloric acid.

i) Write the equation for the reaction of calcium carbonate with hydrochloric acid.

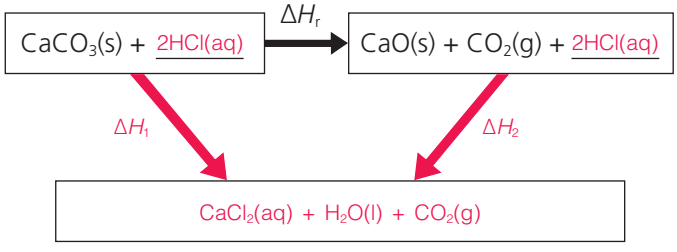
$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$ ΔH_1

(ΔH_1 is the enthalpy change for this reaction)

ii) The reaction of calcium oxide with hydrochloric acid is

$CaO(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l)$ ΔH_2

Use the equations in parts (i) and (ii) to complete the enthalpy change cycle below to show how you could calculate the enthalpy change for the decomposition of calcium carbonate, ΔH_r. Label the arrows in your cycle.



iii) Complete the expression for ΔH_r in terms of ΔH_1 and ΔH_2.

$\Delta H_r =$ $\Delta H_1 - \Delta H_2$

b) Suggest TWO reasons why the value obtained by carrying out these two experiments and using the equation gives a value different to the data booklet value for the decomposition reaction of calcium carbonate.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2012, 21)

15 a) State Hess's Law.

Refer to page T23.

b) Methane burns in a limited supply of oxygen to give carbon monoxide and water.

$$CH_4(g) + \frac{3}{2}O_2(g) \longrightarrow CO(g) + 2H_2O(l)$$

The enthalpy change for this reaction cannot be determined directly but can be found using the standard enthalpy changes of combustion of methane and carbon monoxide, together with Hess's Law.

The standard enthalpy changes of combustion needed are for CH_4, $-890\ kJ\ mol^{-1}$, and for CO, $-283\ kJ\ mol^{-1}$.

i) Draw an enthalpy change cycle which would enable you to calculate the enthalpy change for the combustion of methane to carbon monoxide.

$$CH_4(g) + \frac{3}{2}O_2(g) \longrightarrow CO(g) + 2H_2O(l)$$

ii) Calculate the enthalpy change for this reaction, in $kJ\ mol^{-1}$.

$-607\ kJ\ mol^{-1}$

iii) Explain why the enthalpy change for this reaction cannot be determined directly.

c) Explain why the calculation in part (b)(ii) would give an incorrect result for the enthalpy change for the reaction below.

$$CH_4(g) + \frac{3}{2}O_2(g) \longrightarrow CO(g) + 2H_2O(g)$$

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 23)

14 b) Any two of the following:

- Heat loss to the surroundings.
- Density and specific heat capacity of the solution not being exactly the same as those of water.
- Experiment not carried out under standard conditions.
- Large relative error in temperature measurement.

16 The standard enthalpy change of formation of hexane can be defined as:

'The enthalpy change when 1 mol of hexane is formed from its constituent elements in their standard states under standard conditions'.

Hexane melts at –95 °C and boils at 69 °C.

a) What are 'standard conditions'? The pressure is 1 atmosphere and the temperature is 25 °C.

b) An incomplete equation is shown below for the chemical change that takes place to produce the standard enthalpy change of formation of hexane.

Add state symbols to the equation to show each species in its standard state.

$6C(\text{graphite}) + 7H_2(g) \longrightarrow C_6H_{14}(l)$

c) It is very difficult to determine the standard enthalpy change of formation of hexane directly. Suggest a reason why. Carbon and hydrogen do not react to produce hexane under normal conditions.

d) The standard enthalpy change of formation of hexane can be determined indirectly.
Refer to next page.
Calculate the standard enthalpy change of formation of hexane using the standard enthalpy changes of combustion below.

Substance	$\Delta H_c^\ominus$ (kJ mol^{-1})
C	–394
H_2	–286
C_6H_{14}	–4 163

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2009, 2(c))

17 Hydrazine (N_2H_4) can be used as a rocket fuel. It burns in oxygen to produce nitrogen and water.

$N_2H_4(l) + O_2(g) \longrightarrow N_2(g) + 2H_2O(l)$

Under standard conditions, complete combustion of 1.92 g of hydrazine gives 37.4 kJ of heat.

Calculate the standard enthalpy change of formation of hydrazine.
Refer to next page.
(Relative atomic masses: H = 1.0, N = 14.0; standard enthalpy change of formation of $H_2O(l)$ = –286 kJ mol^{-1})

18 Ethyne (C_2H_2) is formed when water reacts with calcium carbide, CaC_2.

$CaC_2(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(s) + C_2H_2(g)$

The standard enthalpy change of this reaction can be determined indirectly using standard enthalpy changes of formation.
Refer to page T23.

a) What is meant by the term 'standard enthalpy change of formation', $\Delta H_f^\ominus$? You should state the standard conditions in your answer.

b) Standard enthalpy changes of formation are shown in the table below.

Substance	Standard enthalpy change of formation, $\Delta H_f^\ominus$ (kJ mol^{-1})
$CaC_2(s)$	–60
$H_2O(l)$	–286
$Ca(OH)_2(s)$	–987
$C_2H_2(g)$	+227

Calculate the standard enthalpy change of the reaction:
–128 kJ mol^{-1}
$CaC_2(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(s) + C_2H_2(g)$

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2010, 1(d))

19 Upon heating, magnesium nitrate decomposes according to the following equation:

$2Mg(NO_3)_2(s) \longrightarrow 2MgO(s) + 4NO_2(g) + O_2(g)$
$\Delta H_r^\ominus = +538$ kJ

The following $\Delta H_f^\ominus$ values are known:

Substance	$\Delta H_f^\ominus$ (kJ mol^{-1})
MgO(s)	–602
$NO_2(g)$	+39.9

Refer to page T24.

a) Calculate $\Delta H_f^\ominus[Mg(NO_3)_2(s)]$.
–791 kJ mol^{-1}

b) Hence comment on the stability of $Mg(NO_3)_2(s)$ under standard conditions.

Unit Exercise

20 Propanone, C_3H_6O, undergoes complete combustion to form carbon dioxide and water.

$C_3H_6O(l) + 4O_2(g) \longrightarrow 3CO_2(g) + 3H_2O(l)$

Refer to page T24.

a) In an experiment to calculate the enthalpy change of combustion of propanone, 2.90 g of propanone were burnt completely in oxygen.

The heat from this combustion raised the temperature of 200.0 g of water from 20.2 °C to 78.4 °C.

(Specific heat capacity of water = 4.18 J g^{-1} K^{-1})

i) Calculate the number of moles of propanone present in 2.90 g.
0.0500 mol
(Molar mass of propanone = 58.0 g mol^{-1})

ii) Calculate the heat transferred to raise the temperature of 200.0 g of water from 20.2 °C to 78.4 °C.
48.7 kJ

iii) Use your answers to (a)(i) and (ii) to calculate a value for the enthalpy change of combustion of propanone.
–974 kJ mol^{-1}

b) In another experiment, the enthalpy change of combustion of butanone, C_4H_8O, was found to be –1 300 kJ mol^{-1}.

A data book value for the standard enthalpy change of combustion of butanone is –2 440 kJ mol^{-1}.

i) Suggest a reason why the value obtained in the experiment is so different from the data book value.

ii) This data book value (–2 440 kJ mol^{-1}) refers to the following equation.

$$C_4H_8O(l) + \frac{11}{2}O_2(g) \longrightarrow 4CO_2(g) + 4H_2O(l)$$

How would the value be different if it referred to the formation of water in the gaseous state? Justify your answer.

c) Standard enthalpy changes of combustion can be used to calculate the standard enthalpy change of formation of a compound.

i) Define the term 'standard enthalpy change of formation', making clear the meaning of 'standard' in this context.

ii) Use the standard enthalpy changes of combustion, $\Delta H_c^{\ominus}$, given in the table below to find the standard enthalpy change of formation of ethanoic acid, CH_3COOH, in kJ mol^{-1}.
–490 kJ mol^{-1}

Substance	$\Delta H_c^{\ominus}$ (kJ mol^{-1})
C(graphite)	–394
$H_2(g)$	–286
$CH_3COOH(l)$	–870

$2C(graphite) + 2H_2(g) + O_2(g) \longrightarrow CH_3COOH(l)$

(Edexcel Advanced Subsidiary GCE, Unit 1, Jun. 2009, 24(a)–(c))

Unit Exercise

16 d) $\Delta H_f^{\ominus}[compound] = \sum \Delta H_c^{\ominus}[constituent\ elements] - \Delta H_c^{\ominus}[compound]$
$\Delta H_f^{\ominus}[C_6H_{14}(l)] = 6 \times \Delta H_c^{\ominus}[C(graphite)] + 7 \times \Delta H_c^{\ominus}[H_2(g)] - \Delta H_c^{\ominus}[C_6H_{14}(l)]$
$= [6(-394) + 7(-286) - (-4\ 163)]$ kJ mol^{-1}
$= -203$ kJ mol^{-1}

17 Number of moles of hydrazine burnt
$= \frac{1.92\ g}{(2 \times 14.0 + 4 \times 1.0)\ g\ mol^{-1}}$
$= 0.0600$ mol
Standard enthalpy change of combustion of hydrazine
$= \frac{-37.4\ kJ}{0.0600\ mol}$
$= -623$ kJ mol^{-1}
$\Delta H_r^{\ominus} = \sum \Delta H_f^{\ominus}[products] - \sum \Delta H_f^{\ominus}[reactants]$
$\Delta H_r^{\ominus} = 2 \times \Delta H_f^{\ominus}[H_2O(l)] - \Delta H_f^{\ominus}[N_2H_4(l)]$
$\Delta H_f^{\ominus}[N_2H_4(l)] = 2 \times \Delta H_f^{\ominus}[H_2O(l)] - \Delta H_r^{\ominus}$
$= [2(-286) - (-623)]$ kJ mol^{-1}
$= +51$ kJ mol^{-1}
∴ the standard enthalpy change of formation of hydrazine is +51 kJ mol^{-1}.

21 a) At a high temperature, carbon undergoes combustion when it reacts with oxygen.
Refer to page T24.

i) Suggest why it is NOT possible to measure the enthalpy change directly for the following combustion reaction.

$$C(\text{graphite}) + \frac{1}{2}O_2(g) \longrightarrow CO(g)$$

The reaction also forms $CO_2(g)$.

ii) State Hess's Law.

iii) State the meaning of the term 'standard enthalpy change of combustion'.

b) Use the standard enthalpy changes of formation in the table below and the equation to calculate a value for the standard enthalpy change for the extraction of iron using carbon monoxide.

+1 kJ mol^{-1}

Substance	$\Delta H_f^{\ominus}$ (kJ mol^{-1})
$Fe_2O_3(s)$	–822
$CO(g)$	–111
$Fe(l)$	+14
$CO_2(g)$	–394

$$Fe_2O_3(s) + 3CO(g) \longrightarrow 2Fe(l) + 3CO_2(g)$$

c) i) Write an equation for the reaction that corresponds to the standard enthalpy change of formation of carbon dioxide.

ii) State why the value quoted in part (b) for the standard enthalpy change of formation of $CO_2(g)$ is the same as the value for the standard enthalpy change of combustion of carbon.

(AQA Advanced Subsidiary GCE, Unit 2, Jan. 2012, 4(b)–(d))

22 Glucose, $C_6H_{12}O_6$, can be completely combusted to give carbon dioxide and water.

$$C_6H_{12}O_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(l)$$

Refer to page T25.

a) In the body, the conversion of glucose into carbon dioxide and water takes place in a number of stages catalyzed by enzymes.

What name is given to this oxidation process in the body?

Respiration

b) A student carries out an experiment to determine the enthalpy change of combustion of glucose.

In the experiment, 0.831 g of glucose is burnt. The heat released is used to heat 100.0 cm^3 of water from 23.7 °C to 41.0 °C.

i) Calculate the heat released, in kJ, during combustion of 0.831 g of glucose.

7.23 kJ

(Specific heat capacity of water = 4.18 J g^{-1} K^{-1}; density of water = 1.00 g cm^{-3})

ii) Calculate the amount, in moles, of glucose that is burnt.

0.00462 mol

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

iii) Calculate the enthalpy change of combustion of glucose.

–1 560 kJ mol^{-1}

c) The standard enthalpy change of combustion of glucose can also be determined indirectly.

Calculate the standard enthalpy change of combustion of glucose using the standard enthalpy changes of formation below.

Substance	$\Delta H_f^{\ominus}$ (kJ mol^{-1})
$C_6H_{12}O_6(s)$	–1 250
$CO_2(g)$	–394
$H_2O(l)$	–286

–2 830 kJ mol^{-1}

$$C_6H_{12}O_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(l)$$

d) Suggest TWO reasons why enthalpy changes of combustion determined experimentally are less exothermic than the calculated theoretical values.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jan. 2010, 3)

23 The enthalpy change of formation ΔH_f of ZnO(s) can be determined indirectly from the enthalpy change of formation of $H_2O(l)$ and the enthalpy changes of reactions (1) and (2) below.

$ZnO(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2O(l)$ (1)

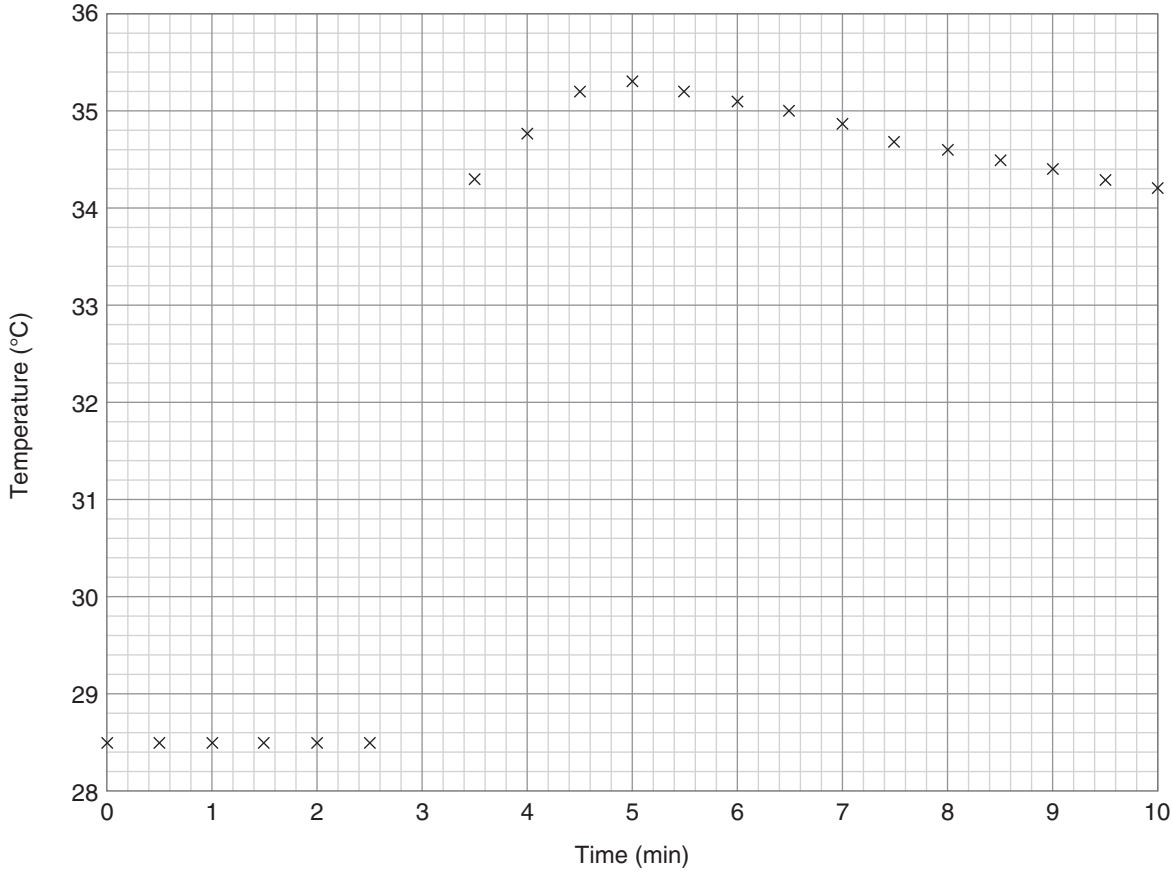
$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$ (2)

An experiment as outlined below was carried out to determine the enthalpy change of reaction (1):

25.0 cm^3 of 1.10 mol dm^{-3} HCl(aq) were placed in an expanded polystyrene cup. The temperature of the acid in the cup was measured with a thermometer at half-minute intervals.

Right at the third minute, 0.75 g of ZnO(s) was added to the cup. The mixture in the cup was then stirred with the thermometer and its temperature was measured for an additional 7 minutes.

The recordings of temperature are shown in the graph below:



a) i) Deduce the greatest temperature change of the reaction mixture. (Show your working on the graph.)

ii) Calculate the molar enthalpy change of reaction (1) under the conditions of the experiment. (Assume that the heat capacity of the expanded polystyrene cup is negligible, and that the specific heat capacity and density of the solutions are 4.2 J g^{-1} K^{-1} and 1.0 g cm^{-3} respectively.)

b) Given that under the same conditions, the molar enthalpy change of reaction (2) is –49 kJ, and the molar enthalpy change of formation of $H_2O(l)$ is –286 kJ, calculate ΔH_f of ZnO(s).

(HKASLE, Paper 2, 2011, 3)

Topic Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Multiple choice questions

1 Which of the following processes is endothermic?

A Freezing of water
B Condensation of steam
C Reaction of $H^+(aq)$ with $OH^-(aq)$ to give $H_2O(l)$
(D) Electrolysis of water

(HKASLE, Paper 1, 2010, 8(a))

2 The enthalpy change of the reaction

$$C(graphite) + \frac{1}{2}O_2(g) \longrightarrow CO(g)$$

cannot be measured directly since some carbon dioxide is always formed in the reaction. It can be calculated using Hess's Law and the enthalpy changes of combustion of graphite and of carbon monoxide.

$$C(graphite) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H = -394 \text{ kJ mol}^{-1}$$

$$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g) \qquad \Delta H = -283 \text{ kJ mol}^{-1}$$

The enthalpy change for the reaction of graphite with oxygen to give carbon monoxide is

Refer to page T26 for explanations.

A -677 kJ mol^{-1}.
B $+111 \text{ kJ mol}^{-1}$.
(C) -111 kJ mol^{-1}.
D $+677 \text{ kJ mol}^{-1}$.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2012, 7)

3 The standard enthalpy changes of formation of carbon dioxide and of methanoic acid are -394 kJ mol^{-1} and -409 kJ mol^{-1} respectively. Calculate the enthalpy change for the reaction

$$H_2(g) + CO_2(g) \longrightarrow HCOOH(l)$$

Refer to next page.

A -803 kJ mol^{-1}
(B) -15 kJ mol^{-1}
C $+803 \text{ kJ mol}^{-1}$
D $+15 \text{ kJ mol}^{-1}$

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 5)

!4 Chlorine dioxide, ClO_2, can be prepared in the laboratory according to the following equation:

$$AgClO_3(s) + \frac{1}{2}Cl_2(g) \longrightarrow AgCl(s) + ClO_2(g) + \frac{1}{2}O_2(g) \qquad \Delta H^{\ominus} = +5 \text{ kJ mol}^{-1}$$

The following values are known:

Substance	$\Delta H_f^{\ominus}$ (kJ mol^{-1})
$AgClO_3(s)$	−30
$AgCl(s)$	−127

What is the standard enthalpy change of formation of $ClO_2(g)$?

Refer to next page.

A -162 kJ mol^{-1}
B -102 kJ mol^{-1}
(C) $+102 \text{ kJ mol}^{-1}$
D $+162 \text{ kJ mol}^{-1}$

5 Consider the standard enthalpy changes of the following reactions:

$$I_2(s) + Cl_2(g) \longrightarrow 2ICl(s) \qquad \Delta H^{\ominus} = +14 \text{ kJ mol}^{-1}$$

$$ICl(s) + Cl_2(g) \longrightarrow ICl_3(s) \qquad \Delta H^{\ominus} = -88 \text{ kJ mol}^{-1}$$

What is the standard enthalpy change of formation of $ICl_3(s)$?

(A) -81 kJ mol^{-1}
B -74 kJ mol^{-1}
C $+74 \text{ kJ mol}^{-1}$
D $+81 \text{ kJ mol}^{-1}$

(HKDSE, Practice paper 1A, 2012, 12)

6 Under standard conditions, complete combustion of 0.050 mol of propane (C_3H_8) gives 111 kJ of heat. Which of the following is the standard enthalpy change of formation of propane?

(Standard enthalpy change of formation of $H_2O(l)$ = –286 kJ mol^{-1};
standard enthalpy change of formation of $CO_2(g)$ = –394 kJ mol^{-1})

(A) –106 kJ mol^{-1}
B +106 kJ mol^{-1}
C –569 kJ mol^{-1}
D +569 kJ mol^{-1}

(HKDSE, Paper 1A, 2013, 18)

7 For which of the following reactions does the value of $\Delta H^{\ominus}$ represent both a standard enthalpy change of combustion and a standard enthalpy change of formation?

(1) $C(graphite) + O_2(g) \longrightarrow CO_2(g)$
(2) $2C(graphite) + O_2(g) \longrightarrow 2CO(g)$
(3) $CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

Directions : Each question (Questions 8–10) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :

A Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C The 1st statement is false but the 2nd statement is true.
D Both statements are false.

	1st statement	2nd statement	
8	Diamond is more stable than graphite.	The conversion of graphite to diamond is an exothermic process.	D
9	The reaction of charcoal with oxygen is endothermic.	Charcoal that is placed in fire can be ignited.	C
10	The standard enthalpy change of neutralization between HCl(aq) and KOH(aq) is the same as that between HNO_3(aq) and KOH(aq).	Both HCl(aq) and HNO_3(aq) are monobasic acids.	B

(HKCEE, Paper 2, 2010, 50)

8 Graphite is more stable than diamond.
The conversion of diamond to graphite is an exothermic process.

9 Both HCl(aq) and HNO_3(aq) are strong acids. The standard enthalpy change of neutralization between a strong acid and a strong alkali is –57.1 kJ mol^{-1}.

3 $\Delta H_r^{\ominus} = \sum \Delta H_f^{\ominus}[products] - \sum \Delta H_f^{\ominus}[reactants]$
$= \Delta H_f^{\ominus}[HCOOH(l)] - \Delta H_f^{\ominus}[CO_2(g)]$
$= [(-409) - (-394)]$ kJ mol^{-1}
$= -15$ kJ mol^{-1}
$\therefore$ the enthalpy change for the reaction is –15 kJ mol^{-1}.

4 $\Delta H_r^{\ominus} = \sum \Delta H_f^{\ominus}[products] - \sum \Delta H_f^{\ominus}[reactants]$
$= \Delta H_f^{\ominus}[AgCl(s)] + \Delta H_f^{\ominus}[ClO_2(g)] - \Delta H_f^{\ominus}[AgClO_3(s)]$
$\Delta H_f^{\ominus}[ClO_2(g)] = \Delta H_r^{\ominus} - \Delta H_f^{\ominus}[AgCl(s)] + \Delta H_f^{\ominus}[AgClO_3(s)]$
$= [(+5) - (-127) + (-30)]$ kJ mol^{-1}
$= +102$ kJ mol^{-1}
$\therefore$ the standard enthalpy change of formation of $ClO_2(g)$ is +102 kJ mol^{-1}.

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

11 Iron powder can be used to make 'warm packs' for keeping users warm. A kind of warm pack is made by putting iron powder in a package which allows air to pass through. The package also contains other substances for speeding up the production of heat.

a) According to the given information, suggest why this kind of warm pack can produce heat.

b) Explain why iron powder, instead of a piece of iron with the same mass, is put in the warm pack.

c) The other substances in the package include moist sodium chloride. Suggest why it can speed up the production of heat.

(HKCEE, Paper 1, 2009, 3(a)–(c))

12 a) Give the meaning of the term 'enthalpy change'.
Refer to page T26.

b) State Hess's Law.

c) Define the term 'standard enthalpy change of formation'.

d) The following data of standard enthalpy change of formation refer to the second stage in the extraction of titanium.

Substance	$\Delta H_f^\ominus$ (kJ mol^{-1})
$TiCl_4(g)$	–720
Na(l)	+3
NaCl(s)	–411
Ti(s)	0

i) State why the value for the standard enthalpy change of formation of Na(l) is NOT zero.

ii) Use data from the table to calculate a value for the standard enthalpy change of the following reaction.

$$TiCl_4(g) + 4Na(l) \longrightarrow 4NaCl(s) + Ti(s)$$

–936 kJ mol^{-1}

iii) State the role of sodium in this reaction.

(AQA Advanced Subsidiary GCE, Unit 2, Jun. 2011, 2(b)–(e)).

13 The table below lists the standard enthalpy changes of formation of four compounds:

Compound	$\Delta H_{f,\,298}^\ominus$ (kJ mol^{-1})
$H_2O(l)$	–286
$H_2O_2(l)$	–188
$NH_3(g)$	–46
$N_2H_4(l)$	+51

a) What is the meaning of the term 'standard enthalpy change of formation'?

b) Hydrazine (N_2H_4) is a colourless liquid commonly used as a rocket fuel. It can be synthesized in a chemical process in which ammonia is oxidized by hydrogen peroxide to give hydrazine and water.

For the oxidation of ammonia to hydrazine,

i) write its chemical equation, and

ii) calculate its standard enthalpy change using the above thermochemical data.

c) A student found the following information in a Material Safety Data Sheet (MSDS):

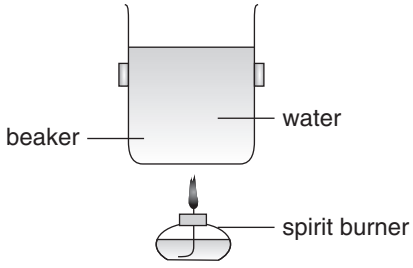
'Hydrazine is extremely explosive in the presence of oxidizing materials.'

The student accounted for the phenomenon by the positive standard enthalpy change of formation of hydrazine. Is the student's explanation correct? Elaborate your answer.

(HKASLE, Paper 2, 2010, 7)

14 The enthalpy change of combustion of ethanol was determined using the apparatus shown in the diagram below. In the experiment, the temperature increase of the water in the beaker was measured when a known mass of the ethanol was burnt.

Refer to page T26.



a) The results of the experiment are summarized in the table below.

Mass of water in the beaker	250.00 g
Mass of spirit burner + contents (initial)	63.21 g
Mass of spirit burner + contents (final)	62.47 g
Temperature of water (initial)	21.0 °C
Temperature of water (final)	31.5 °C

i) Calculate the heat produced by the combustion of the alcohol using the equation.

heat produced (J)
= mass of water x 4.18 x temperature change
11 000 J

ii) Calculate the number of moles of ethanol burnt in this experiment (the formula of ethanol is C_2H_5OH).
0.0160 mol
(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

iii) Calculate the enthalpy change of combustion of ethanol in kJ mol^{-1}.
–688 kJ mol^{-1}

b) The data book value for the enthalpy change of combustion of ethanol is –1 368 kJ mol^{-1}.

i) List THREE ways in which the design of the experiment causes the results to be so different from the data book value.

ii) Use the data book values for enthalpy changes of combustion given in the table below to calculate the enthalpy change of formation of ethanol.
–278 kJ mol^{-1}

Substance	ΔH_c (kJ mol^{-1})
C(graphite)	–394
$H_2(g)$	–286
$C_2H_5OH(l)$	–1 368

(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2011, 19(a)–(b))

15 An experiment was carried out to determine the enthalpy change of the hydrolysis of silicon tetrachloride.

$$SiCl_4(l) + 2H_2O(l) \longrightarrow SiO_2(s) + 4HCl(g)$$

Procedure:

1 Weigh 6.81 g of silicon tetrachloride.

2 Transfer 100 cm^3 of water into a polystyrene cup and record the temperature of the water.

3 Add the silicon tetrachloride to the water.

4 Stir the mixture and record the maximum temperature reached.

The following results were obtained:

Mass of mixture = 107 g
Initial temperature = 25.6 °C
Highest temperature reached = 31.6 °C
Specific heat capacity of the mixture = 4.10 J g^{-1} K^{-1}

Refer to page T27.

a) Assuming that 100 cm^3 of water are in large excess, calculate the enthalpy change when one mole of silicon tetrachloride undergoes hydrolysis.
–65.8 kJ mol^{-1}
(Molar mass of $SiCl_4$ = 170.1 g mol^{-1})

b) In reactions of this type, heat is lost while the reaction is still going on.

Suggest how, using the same apparatus and a stop watch, the procedure could be modified to find the maximum temperature change more accurately.

16 This question is about hexane, C_6H_{14}, and hex-1-ene, C_6H_{12}.

Refer to page T28.

a) The enthalpy change of combustion of hexane was measured using a spirit burner to heat a known mass of water in a calorimeter. The temperature rise of the water was measured. The results of the experiment are shown below.

Mass of hexane burnt	0.320 g
Mass of water in calorimeter	50.0 g
Initial temperature of water	22.0 °C
Final temperature of water	68.0 °C

The specific heat capacity of water is 4.18 J g^{-1} K^{-1}.

i) Calculate the heat in joules produced by burning the hexane.

9 610 J

ii) Calculate the enthalpy change of combustion of hexane. The mass of 1 mole of hexane is 86.0 g.

–2 580 kJ mol^{-1}

iii) The value for the enthalpy change of combustion in this experiment is different from the value given in data books. Suggest TWO reasons for this difference.

iv) A student suggested that the results would be more accurate if a thermometer which read to 0.01 °C was used. Explain why this would NOT improve the accuracy of the result. A calculation is NOT required.

b) Hex-1-ene can be converted to hexane in the following reaction.

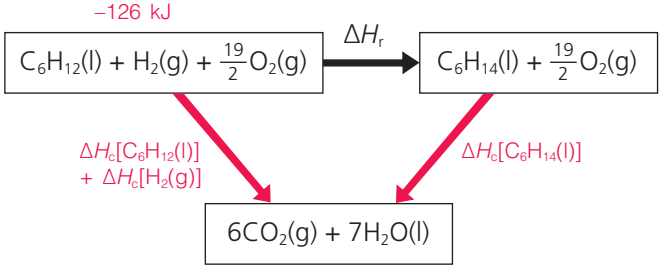
$C_6H_{12}(l) + H_2(g) \longrightarrow C_6H_{14}(l)$

The enthalpy change of this reaction ΔH_r can be calculated from the following enthalpy changes of combustion.

Substance	ΔH_c (kJ mol^{-1})
Hex-1-ene, C_6H_{12}	–4 003
Hydrogen, H_2	–286
Hexane, C_6H_{14}	–4 163

Complete the enthalpy change cycle by adding labelled arrows. Use your cycle to calculate the enthalpy change ΔH_r.

–126 kJ



(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2010, 18(c)–(d))

17 Potassium hydrogencarbonate ($KHCO_3$) can be used to bake bread. Upon heating, $KHCO_3$ decomposes into K_2CO_3, H_2O and CO_2.

a) Explain the purpose of using $KHCO_3$ in bread baking.

b) Write the chemical equation for the decomposition of $KHCO_3$ upon heating.

c) The enthalpy change of decomposition of $KHCO_3(s)$ can be determined indirectly from the enthalpy changes of the following two reactions:

$KHCO_3(s) + HCl(aq) \longrightarrow KCl(aq) + H_2O(l) + CO_2(g)$ *Reaction 1*

$K_2CO_3(s) + 2HCl(aq) \longrightarrow 2KCl(aq) + H_2O(l) + CO_2(g)$ *Reaction 2*

In an experiment to determine the enthalpy change of *Reaction 1*, 3.39 g of $KHCO_3(s)$ were added to excess HCl(aq) in an expanded polystyrene cup. The experimental data obtained are shown below:

Initial temperature of the reacting solution	25.8 °C
Final temperature of the reacting solution	20.2 °C
Mass of the resulting solution	27.5 g
Specific heat capacity of the contents	4.3 J g^{-1} K^{-1}
Molar mass of $KHCO_3$	100.1 g mol^{-1}

i) Assuming that the heat capacity of the cup used is negligible, calculate the enthalpy change of *Reaction 1* from the above data.

ii) In another experiment performed under the same conditions, the enthalpy change of *Reaction 2* was found to be –49.1 kJ mol^{-1}. Calculate the enthalpy change of decomposition of $KHCO_3(s)$ under the experimental conditions.

d) According to the literature, the standard enthalpy changes of formation of $K_2CO_3(s)$, $KHCO_3(s)$, $CO_2(g)$ and $H_2O(l)$ are as follows:

Compound	$\Delta H^{\ominus}_{f,\,298}$ (kJ mol^{-1})
$K_2CO_3(s)$	–1 146
$KHCO_3(s)$	–959
$CO_2(g)$	–394
$H_2O(l)$	–286

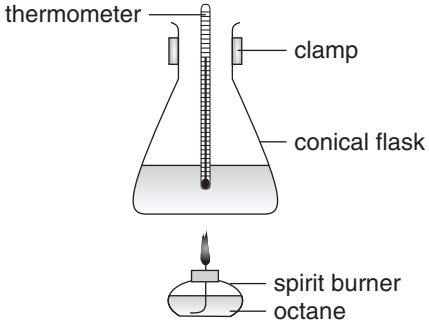
i) Using the given information, calculate the standard enthalpy change of decomposition of $KHCO_3(s)$.

ii) Suggest why the answers obtained from (c)(ii) and (d)(i) are different.

(HKDSE, Paper 1B, 2012, 8)

18 Octane is a component of petrol.
Refer to page T28.

a) A value for the enthalpy change of combustion of octane can be obtained from a simple experiment using the following apparatus.



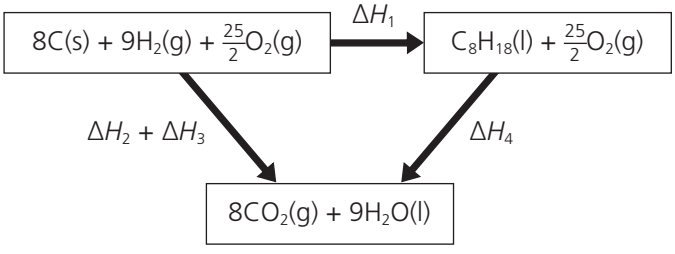
i) Describe how you could use this apparatus to obtain data from which you could calculate a value for the enthalpy change of combustion of octane. You should also explain how you would use your experimental results to calculate this value.

ii) Results obtained using this apparatus give a much less negative value for the enthalpy change of combustion of octane than the data book value.

Suggest TWO limitations in the practical procedure which would result in a less negative value for the enthalpy change of combustion of octane than the data book value.

b) ΔH_f values can be calculated for hydrocarbons using enthalpy changes of combustion in an enthalpy change cycle.

The enthalpy change cycle below can be used to calculate the standard enthalpy change of formation of octane.



i) State the enthalpy changes represented by ΔH_1, ΔH_2, ΔH_3, and ΔH_4.

ii) You are given the following values:

$\Delta H_2 + \Delta H_3 = -5\,718$ kJ mol^{-1};

$\Delta H_4 = -5\,470$ kJ mol^{-1};

Calculate a value for ΔH_1.
–248 kJ mol^{-1}

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F331, Jan. 2010, 1(b)–(c))

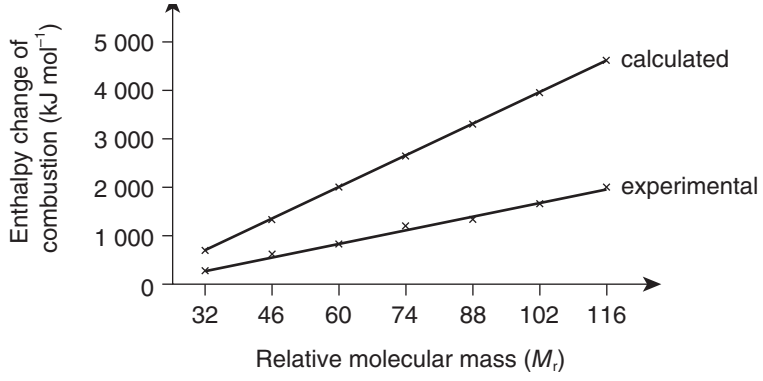
19 a) i) Give the meaning of the term 'standard enthalpy change of combustion'.
Refer to page T29.

ii) Use the standard enthalpy change of formation data from the table and the equation for the combustion of butan-1-ol to calculate a value for the standard enthalpy change of combustion of butan-1-ol.
–2 680 kJ mol^{-1}

Substance	$\Delta H_f^{\ominus}$ (kJ mol^{-1})
$CH_3CH_2CH_2CH_2OH(l)$	–327
$O_2(g)$	0
$CO_2(g)$	–394
$H_2O(l)$	–286

$CH_3CH_2CH_2CH_2OH(l) + 6O_2(g) \longrightarrow 4CO_2(g) + 5H_2O(l)$

b) A student performed an experiment and obtained an experimental value for the enthalpy change of combustion for each alkanol in the alkanol series. These experimental values were then compared with calculated values from standard enthalpy change of formation, as shown in the graph below.



i) In terms of bonds broken and bonds formed, explain why the calculated values of enthalpy changes of combustion of these alkanols, when plotted against M_r, follow a straight line.

ii) Give TWO reasons why the experimental values obtained by the student are lower than the calculated values using the enthalpy change of formation data.

(AQA Advanced Subsidiary GCE, Unit 2, Jan. 2011, 9(c)–(d))

20 Enthalpy changes can be determined directly or indirectly.

a) A student investigates the reaction between magnesium and dilute hydrochloric acid.
Refer to next page.
$Mg(s) + 2HCl(g) \longrightarrow MgCl_2(aq) + H_2(g)$

The student determines the enthalpy change for this reaction.

In her experiment, she reacts 0.486 g of magnesium with 50.0 cm^3 of 2.00 mol dm^{-3} HCl(aq). The HCl(aq) is in excess. The temperature of the solution changes from 19.2 °C to 32.0 °C.

i) Calculate the heat released, in kJ, during this reaction.

(Specific heat capacity of the solution = 4.18 J g^{-1} K^{-1}; density of the solution = 1.00 g cm^{-3})

ii) Calculate the amount, in moles, of magnesium used by the student.

(Relative atomic mass: Mg = 24.3)

iii) Calculate the enthalpy change of reaction.

b) The student wants to determine the enthalpy change of formation of calcium carbonate, $CaCO_3(s)$.

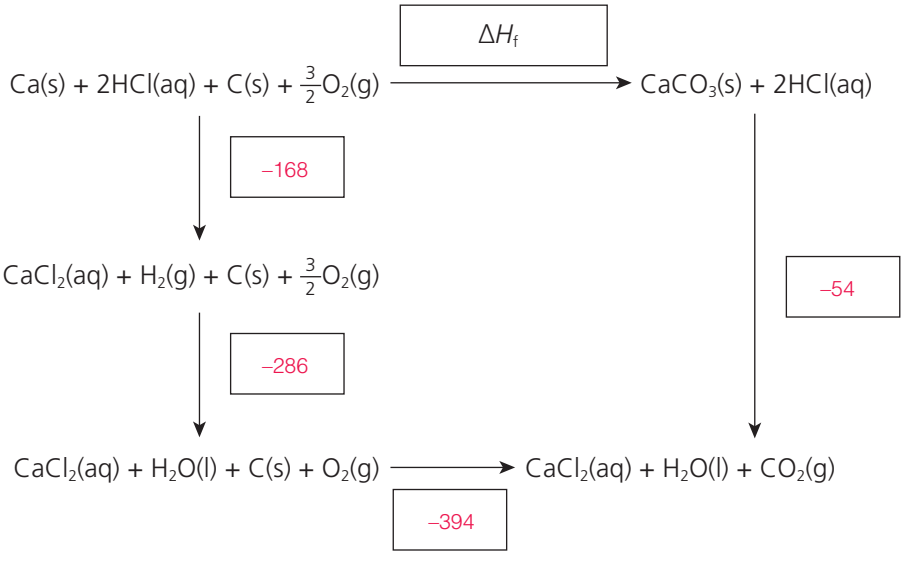
$$Ca(s) + C(s) + \frac{3}{2}O_2(g) \longrightarrow CaCO_3(s)$$

i) What is meant by the term 'standard enthalpy change of formation'? You should state the standard conditions in your answer. The standard enthalpy change of formation of a substance, $\Delta H_f^{\ominus}$, is the enthalpy change when one mole of the substance is formed from its elements in their standard states. The temperature and pressure under the standard conditions are 25 °C and 1 atmosphere respectively.

ii) Using the following data and enthalpy change cycle,

- fill in the boxes on the enthalpy change cycle with the correct enthalpy change values;
- calculate the enthalpy change of formation, ΔH_f, of $CaCO_3(s)$. $\Delta H[CaCO_3(s)] = [-168 - 286 - 394 - (-54)]$ kJ mol^{-1} $= -794$ kJ mol^{-1}

Reaction	ΔH (kJ mol^{-1})
$C(s) + O_2(g) \longrightarrow CO_2(g)$	–394
$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$	–286
$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$	–54
$Ca(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2(g)$	–168



(OCR Advanced Subsidiary GCE, Chem. A, F322, Jan. 2012, 3)

20 a) i) Heat released = 50.0 g x 4.18 J g^{-1} K^{-1} x (32.0 – 19.2) K
= 2 680 J
= 2.68 kJ

ii) Number of moles of magnesium used = $\frac{0.486 \text{ g}}{24.3 \text{ g mol}^{-1}}$
= 0.0200 mol

iii) Enthalpy change of reaction = $\frac{-2.68 \text{ kJ}}{0.0200 \text{ mol}^{-1}}$
= –134 kJ mol^{-1}

21 Magnesium carbonate reacts with dilute nitric acid as shown in the equation below.

$MgCO_3(s) + 2HNO_3(aq) \longrightarrow Mg(NO_3)_2(aq) + CO_2(g) + H_2O(l)$

The enthalpy change for this reaction can be determined as follows:

Procedure

1 Weigh 3.50 g of finely powdered magnesium carbonate.

2 Transfer 50.0 cm^3 of 2.00 mol dm^{-3} nitric acid into a polystyrene cup and record the temperature of the acid.

3 Add the magnesium carbonate to the nitric acid.

4 Stir the mixture and record the maximum temperature reached.

Results

Temperature of nitric acid before addition of magnesium carbonate	21.0 °C
Final temperature of solution	29.7 °C

a) Explain why the magnesium carbonate used in this experiment should be finely powdered rather than in lumps.

b) i) Calculate the number of moles of magnesium carbonate in 3.50 g.

(Assume the molar mass of magnesium carbonate is 84.0 g mol^{-1}.)

ii) The volume of dilute nitric acid used contained 0.100 mole of HNO_3. Suggest why this amount is suitable.

c) i) Calculate the heat transferred, in joules, in this reaction between magnesium carbonate and nitric acid.

(Assume the mass of solution is 50.0 g and its specific heat capacity is 4.18 J g^{-1} K^{-1}.)

ii) Calculate the enthalpy change, ΔH, for the reaction of one mole of magnesium carbonate with nitric acid. Your answer should be in units of kJ mol^{-1}.

d) The nitric acid for this experiment could be measured using either a pipette or a measuring cylinder. Give ONE practical advantage of using each piece of apparatus.

e) State and explain the effect, if any, on the calculated enthalpy change, ΔH, if

i) a copper calorimeter were used instead of the polystyrene cup;

ii) 3.50 g of damp magnesium carbonate were used.

a) Shorten the time for the heat loss to the surroundings during the reaction.
(The use of finely powdered magnesium carbonate increases the rate of reaction and shortens the time needed for completing the reaction. This shortens the time for the heat loss to the surroundings during the reaction and the temperature measured could be more accurate.)

b) i) Number of moles of magnesium carbonate reacted
$= \frac{3.50 \text{ g}}{84.0 \text{ g mol}^{-1}}$
$= 0.0417$ kJ mol
ii) Excess HNO_3 ensures all the magnesium carbonate reacts.

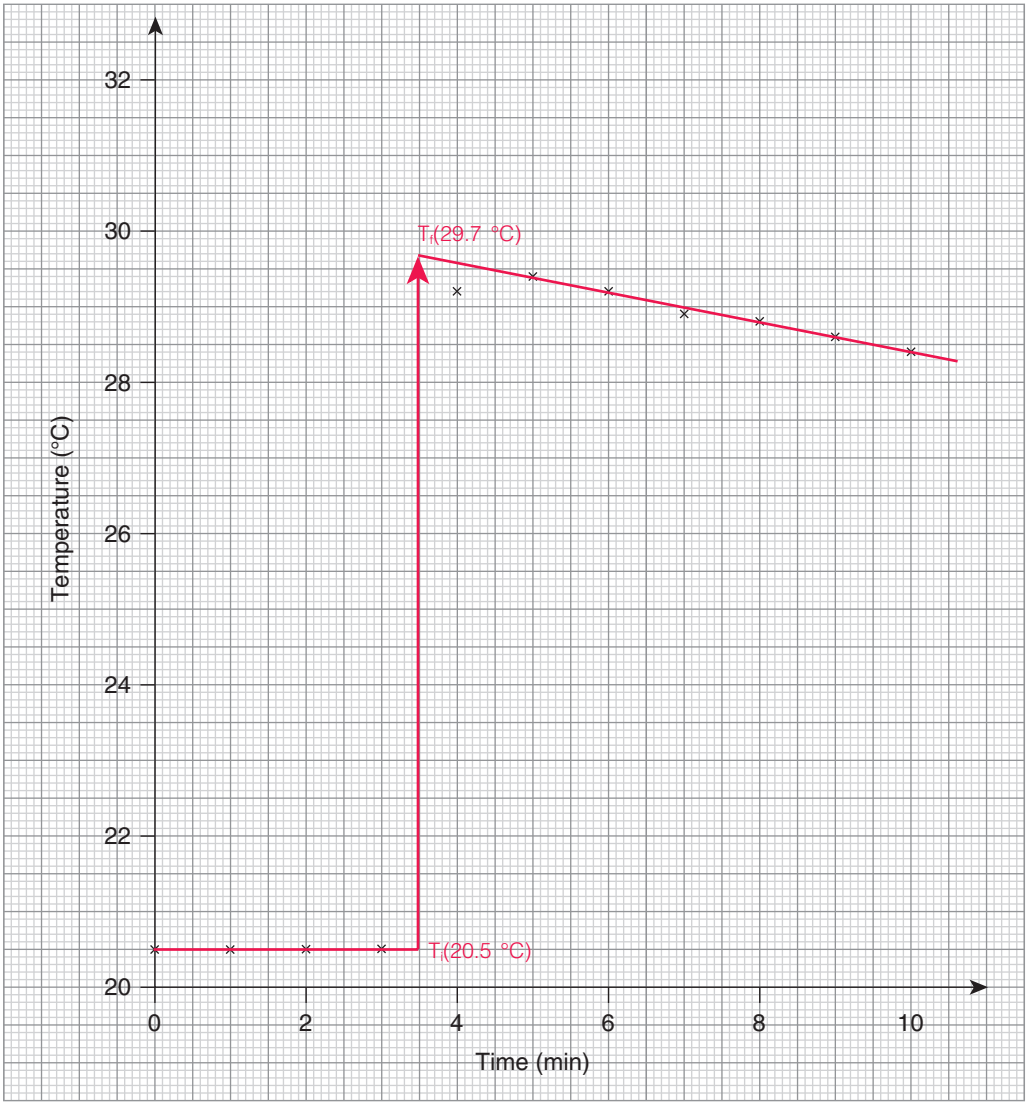
c) i) Heat transferred
$= 50.0 \text{ g} \times 4.18 \text{ J g}^{-1} \text{ K}^{-1} \times (29.7 - 21.0) \text{ K}$
$= 1\,820$ J
ii) $\Delta H = \frac{-1\,820 \text{ J}}{0.0417 \text{ mol}}$
$= -43.6$ kJ mol^{-1}

d) Pipette: more accurate
Measuring cylinder: faster / easier / more convenient

e) i) The calculated enthalpy change is less negative / less exothermic.
Temperature rise lower than expected due to heat loss.
ii) The calculated enthalpy change is less negative / less exothermic.
Less magnesium carbonate undergoes reaction so that the temperature rise is lower.

f) The experiment was repeated with a change in the procedure. The temperature of the dilute nitric acid was measured every minute for three minutes. After exactly three and a half minutes, the magnesium carbonate was added and the mixture was stirred. The temperature of the mixture was then taken every minute for another six minutes.

A graph of the temperature readings against time using this procedure is shown below.



i) Use this graph to calculate the maximum temperature change. Show your working on the graph.
Maximum temperature change = (29.7 – 20.5) °C = 9.2 °C

ii) Why does this method give a more accurate result than the original procedure?
This method performs cooling correction to compensate the heat loss.

(Edexcel Advanced Subsidiary GCE, Unit 3B, Jan. 2012, 3)